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(54) **STORAGE DEVICE WITH VARIABLE CELL AREA SIZE AND OPERATING METHOD THEREOF**

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See application file for complete search history.

(56) **References Cited**  
U.S. PATENT DOCUMENTS

11,237,769 B2 2/2022 Kanno

11,269,558 B2 3/2022 Kanno et al.

11,327,681 B2 5/2022 Shin et al.

11,455,244 B2 9/2022 Tikoo et al.

11,586,385 B1 \* 2/2023 Lercari ..... G06F 1/30

2019/0332298 A1 \* 10/2019 Madabhushi ..... G06F 3/0679

2021/0318820 A1 10/2021 Jin et al.

2022/0137840 A1 5/2022 Lee

2022/0350530 A1 11/2022 Yoshida et al.

2023/0384966 A1 \* 11/2023 Sela ..... G06F 3/0685

2024/0053900 A1 \* 2/2024 Vaghasiya ..... G06F 3/0616

2024/0069806 A1 \* 2/2024 Bolisetty ..... G06F 3/064

2024/0220138 A1 \* 7/2024 Inbar ..... G06F 11/004

\* cited by examiner

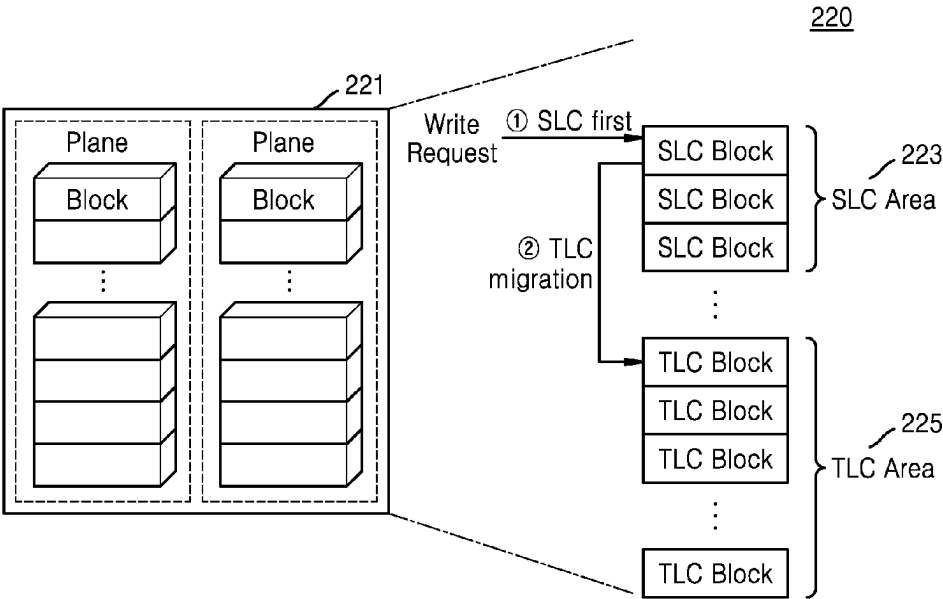
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(57) **ABSTRACT**

A storage device and an operating method of the same are disclosed. The storage device includes a memory controller and a non-volatile memory, wherein the memory controller is configured to calculate a number of zones, which are spaces allocated to logical addresses, based on a cell area size of first storage areas of the non-volatile memory, and provide the number of zones to a host, is configured to generate multiple zones of a number within the calculated number of zones, and to map a logical address of each zone to a physical address of at least one of the first storage areas, and is configured to, in response to a first write request from the host, write data included in the first write request into first storage areas corresponding to a zone indicated by the first write request.

20 Claims, 17 Drawing Sheets



The diagram illustrates a storage device architecture. On the left, a dashed box labeled 221 contains two vertical columns, each labeled "Plane". Each plane contains a stack of blocks, with the top block labeled "Block". Below the top block in each plane are three more blocks, and vertical ellipsis dots indicate further blocks. To the right of the planes, a "Write Request" arrow points to a stack of blocks labeled 223, which are grouped by a bracket and labeled "SLC Area". This stack contains three "SLC Block" units. Below this, another stack of blocks labeled 225 is grouped by a bracket and labeled "TLC Area". This stack contains three "TLC Block" units. A "TLC migration" arrow points from the SLC Area to the TLC Area. Below the TLC Area, there is a single "TLC Block" unit. Vertical ellipsis dots are shown between the SLC Area and the TLC Area, and below the final TLC Block.

FIG. 1

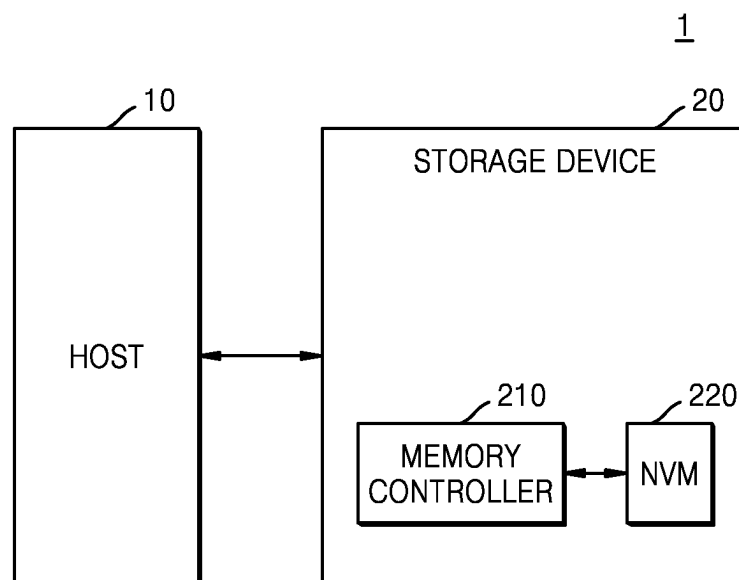


FIG. 2

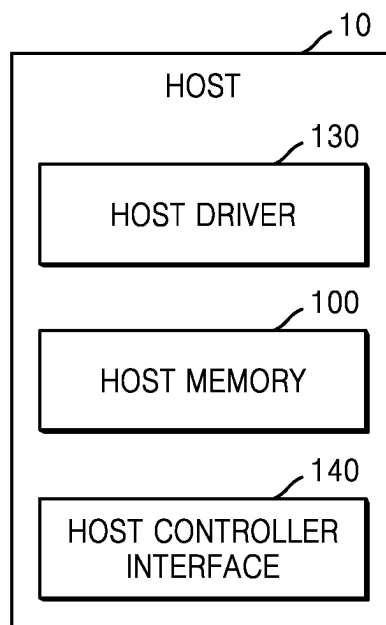


FIG. 3

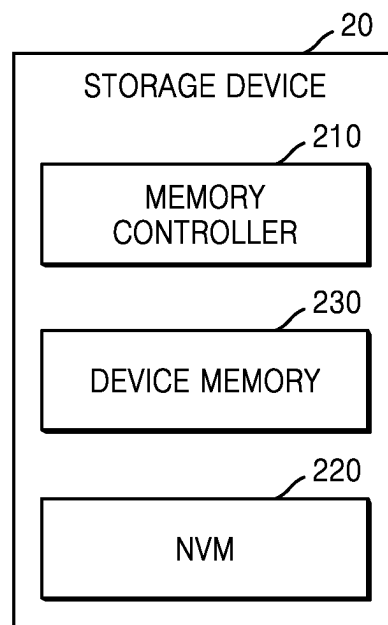


FIG. 4

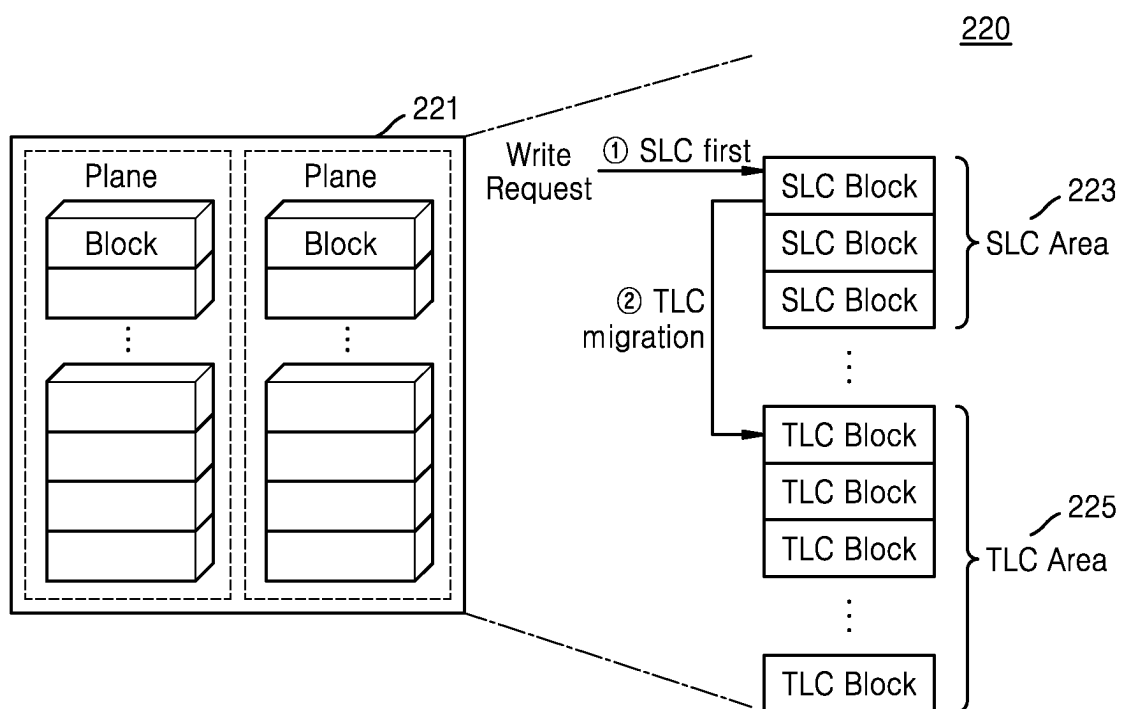


FIG. 5

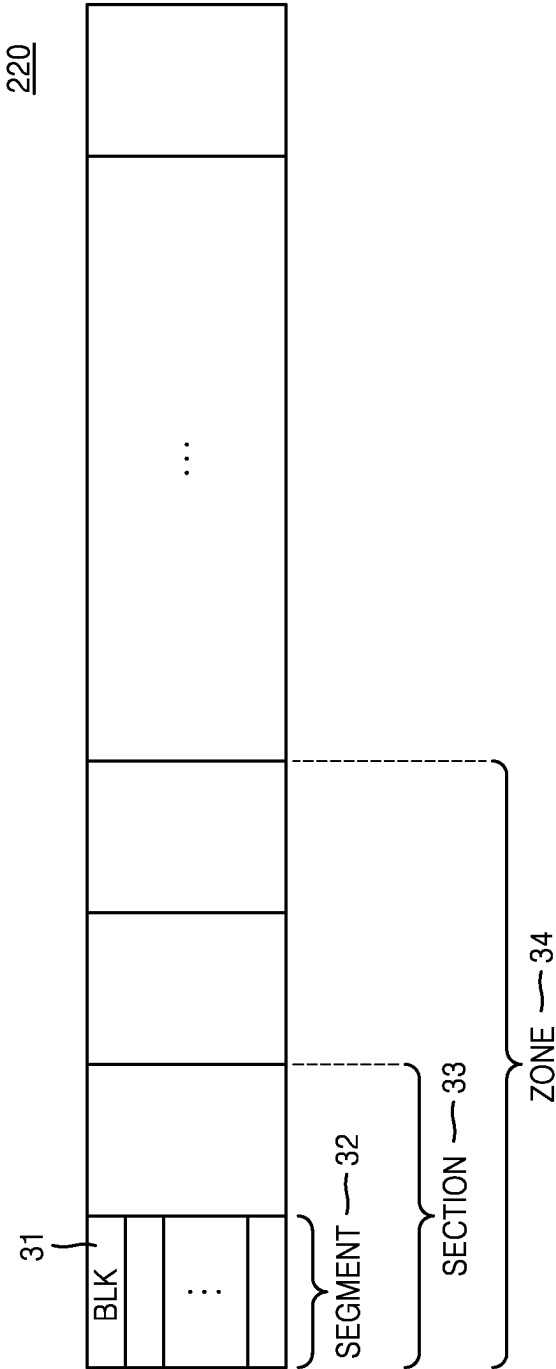


FIG. 6

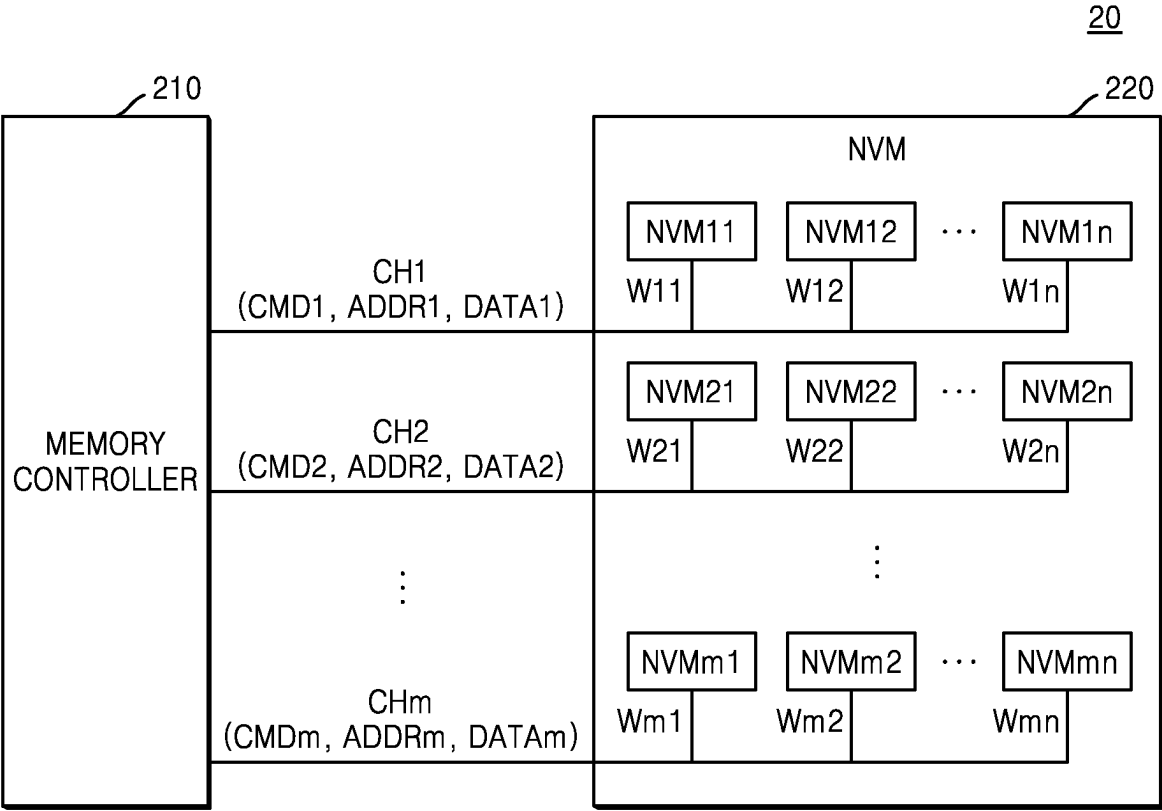


FIG. 7

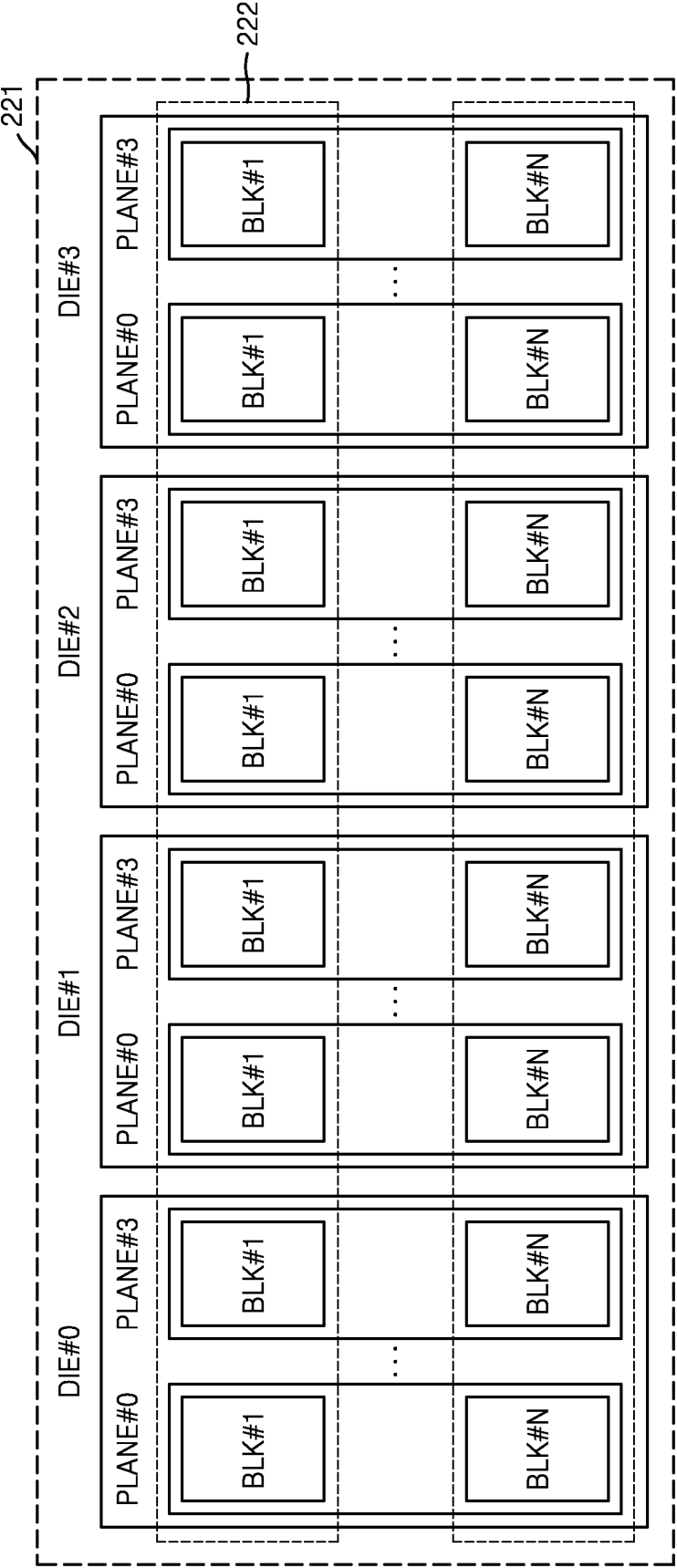




FIG. 8

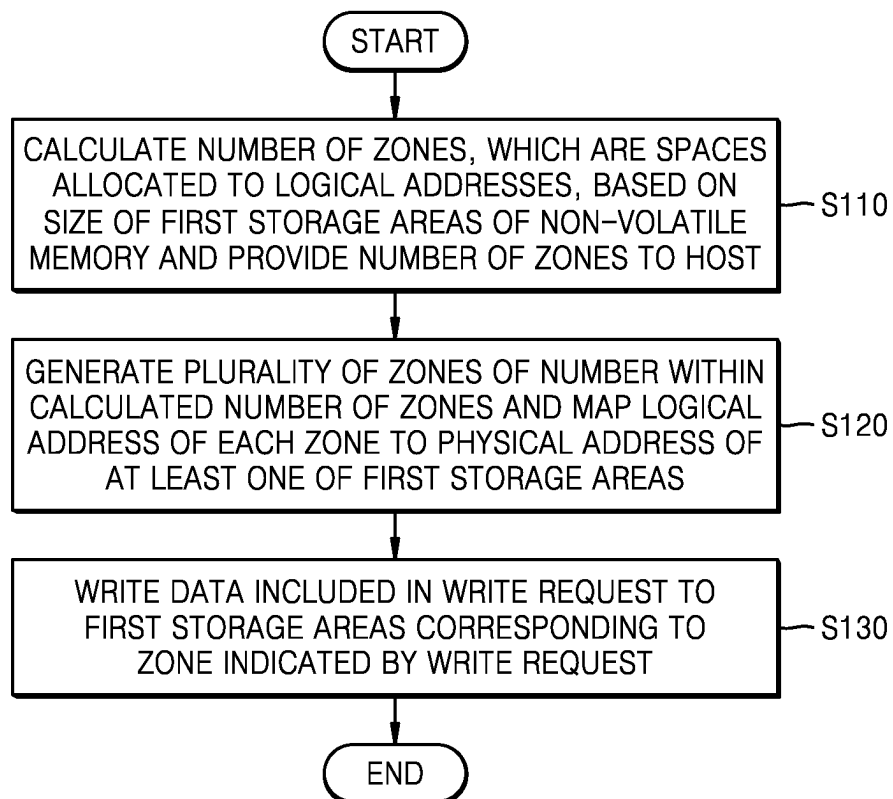


FIG. 9

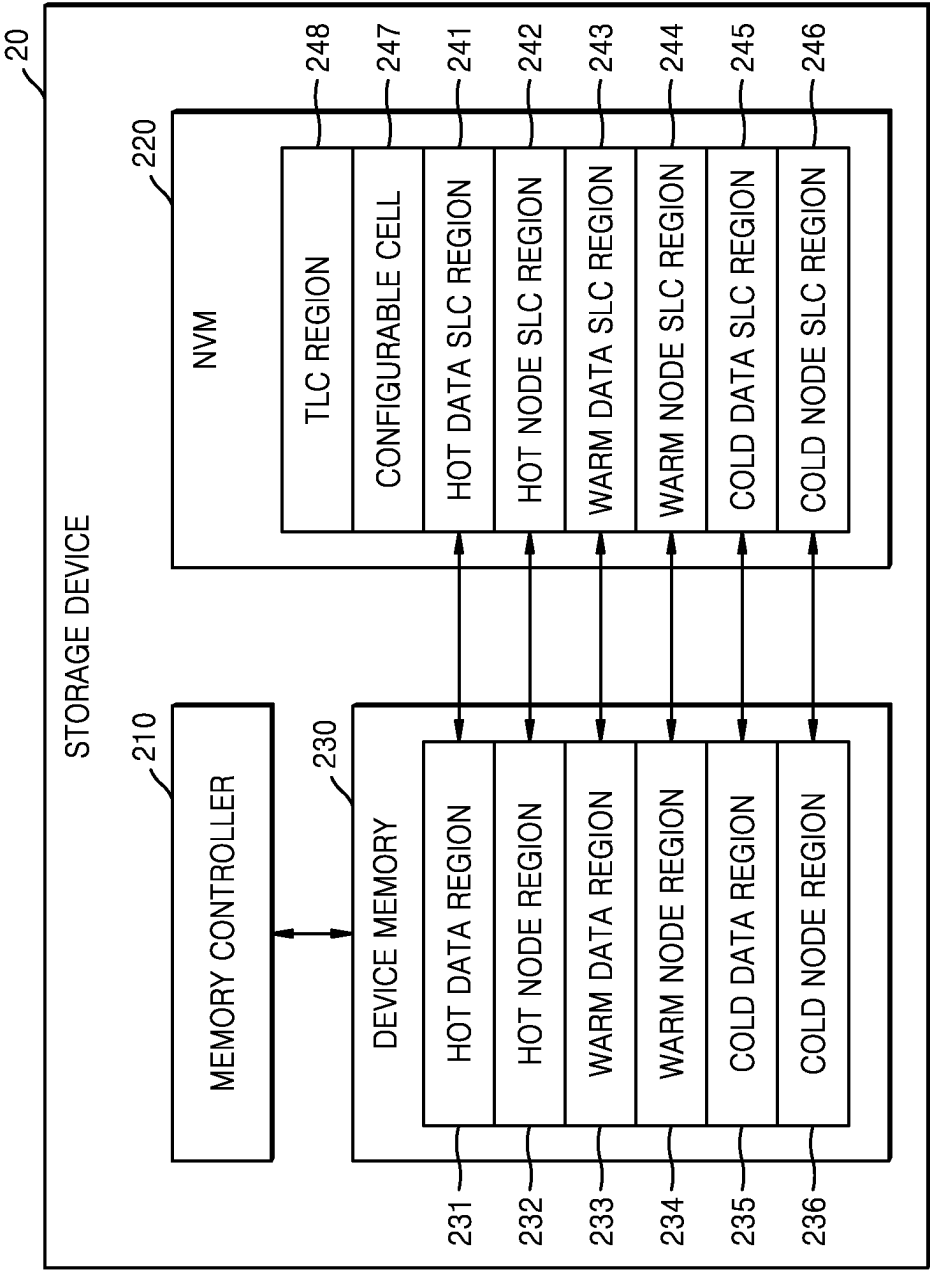


FIG. 10

| General UPIU |                  |                  |                             |                                          |
|--------------|------------------|------------------|-----------------------------|------------------------------------------|
| 0            | Transaction Type |                  | 1                           | Flags                                    |
|              |                  |                  | 2                           | LUN                                      |
| 3            |                  |                  | 3                           | Task Tag                                 |
| 4            | Initiator ID     | Command Set Type | 5                           | Query Function, Task Management Function |
| 6            |                  |                  | 6                           | Response                                 |
| 7            |                  |                  | 7                           | Status                                   |
| 8            | Total EHS Length |                  | 9                           | Device Information                       |
| 10           |                  |                  | 10                          | Data Segment Length                      |
| 11           |                  |                  | 11                          |                                          |
| 12~31        |                  |                  | Transaction Specific Fields |                                          |
| k~j+3        |                  |                  | Extra Header Segment        |                                          |
|              |                  |                  | Header E2ECRC               |                                          |
|              |                  |                  | Data Segment                |                                          |
|              |                  |                  | Data E2ECRC                 |                                          |

FIG. 11

| QUERY REQUEST UPIU           |                        |             |                             |                     |
|------------------------------|------------------------|-------------|-----------------------------|---------------------|
| 0                            | xx01 0110b             | 1           | Flags                       | 23                  |
|                              |                        |             | Reserved                    | Task Tag            |
| 4                            | Reserved               | 5           | Query Function              | 7                   |
|                              |                        |             | Reserved                    | Reserved            |
| 8                            | Total EHS Length (OOH) | 9           | SLC Area Length             | 11 (LSB)            |
|                              |                        |             | (MSB)                       | Data Segment Length |
| 12                           |                        | 13          | Transaction Specific Fields | 15                  |
|                              |                        |             | Transaction Specific Fields |                     |
| 16                           |                        | 17          | Transaction Specific Fields | 19                  |
|                              |                        |             | Transaction Specific Fields |                     |
| 20                           |                        | 21          | Transaction Specific Fields | 23                  |
|                              |                        |             | Transaction Specific Fields |                     |
| 24                           |                        | 25          | Transaction Specific Fields | 27                  |
|                              |                        |             | Transaction Specific Fields |                     |
| 28                           |                        | 29          | Reserved                    | 31                  |
|                              |                        |             | Reserved                    |                     |
| Header E2ECRC (omit if HD=0) |                        |             |                             |                     |
| k                            | Data[0]                | k+1         | Data[1]                     | k+3                 |
|                              |                        |             | Data[2]                     | Data[3]             |
|                              | ...                    |             | ...                         | ...                 |
| k+ Length-4                  | Data[Length-4]         | k+ Length-3 | Data[Length-2]              | k+ Length-1         |
|                              |                        |             | Data[Length-1]              | Data[Length-1]      |
| Data E2ECRC (omit if DD=0)   |                        |             |                             |                     |



FIG. 13

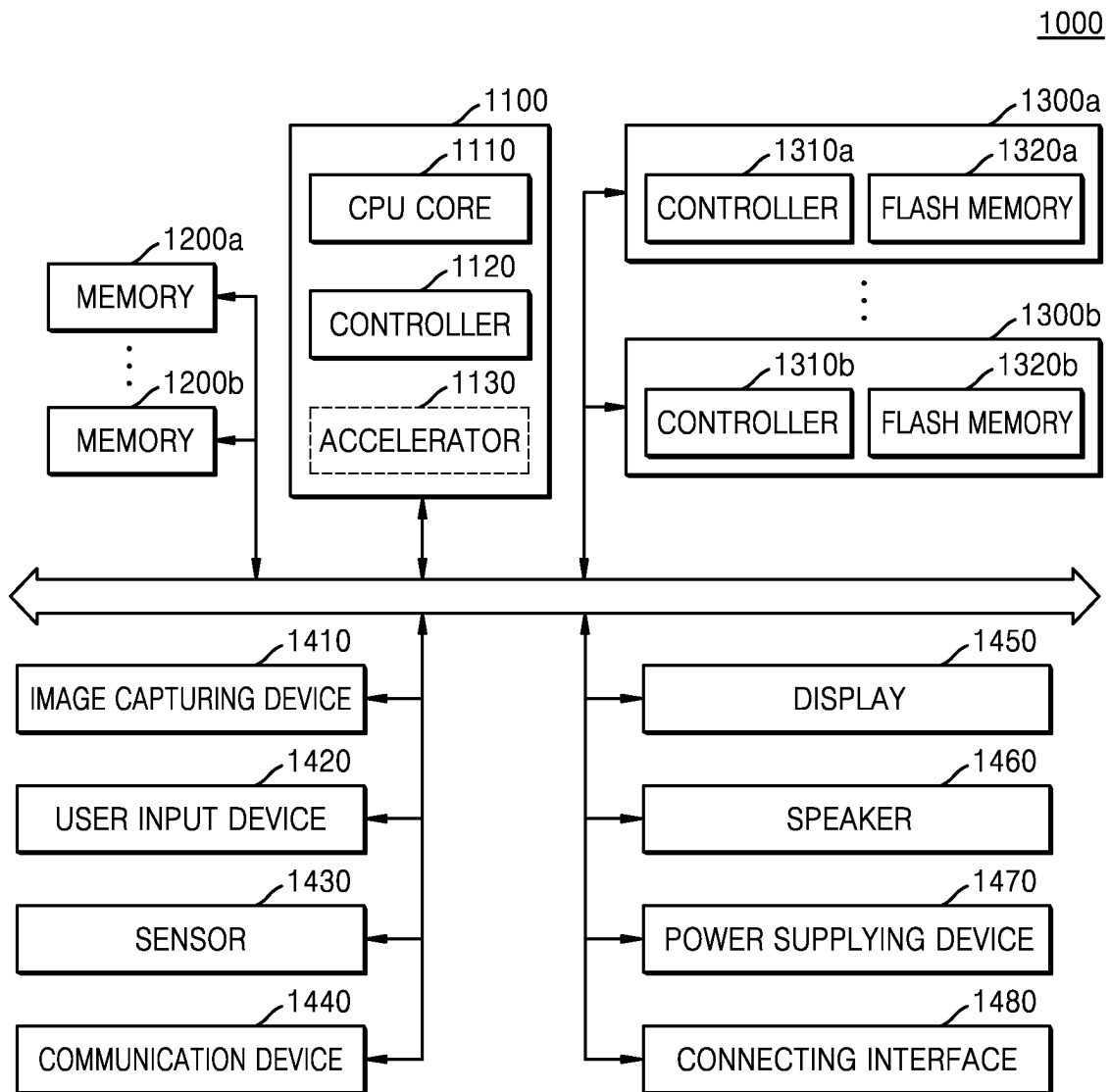


FIG. 14

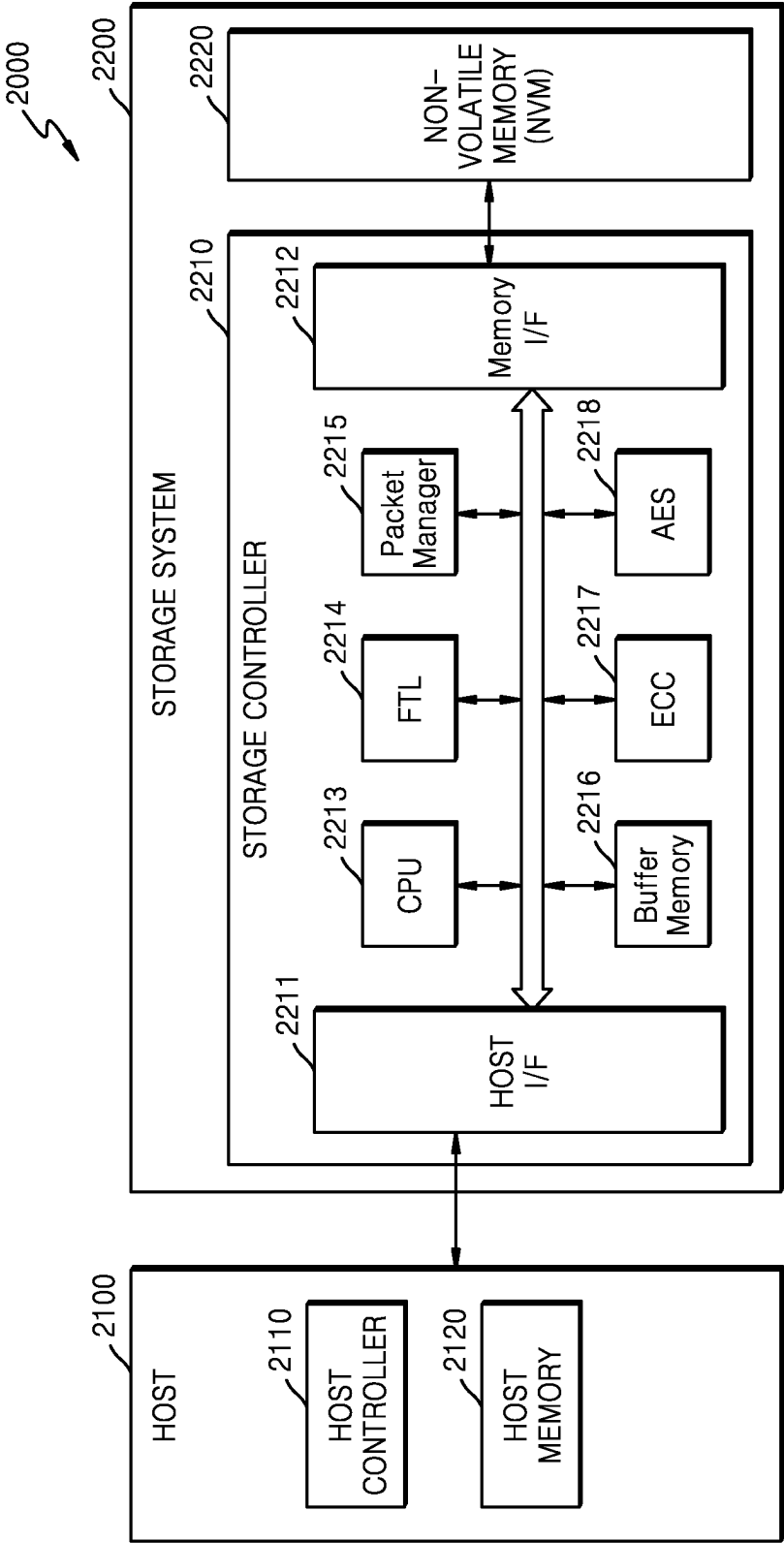


FIG. 15

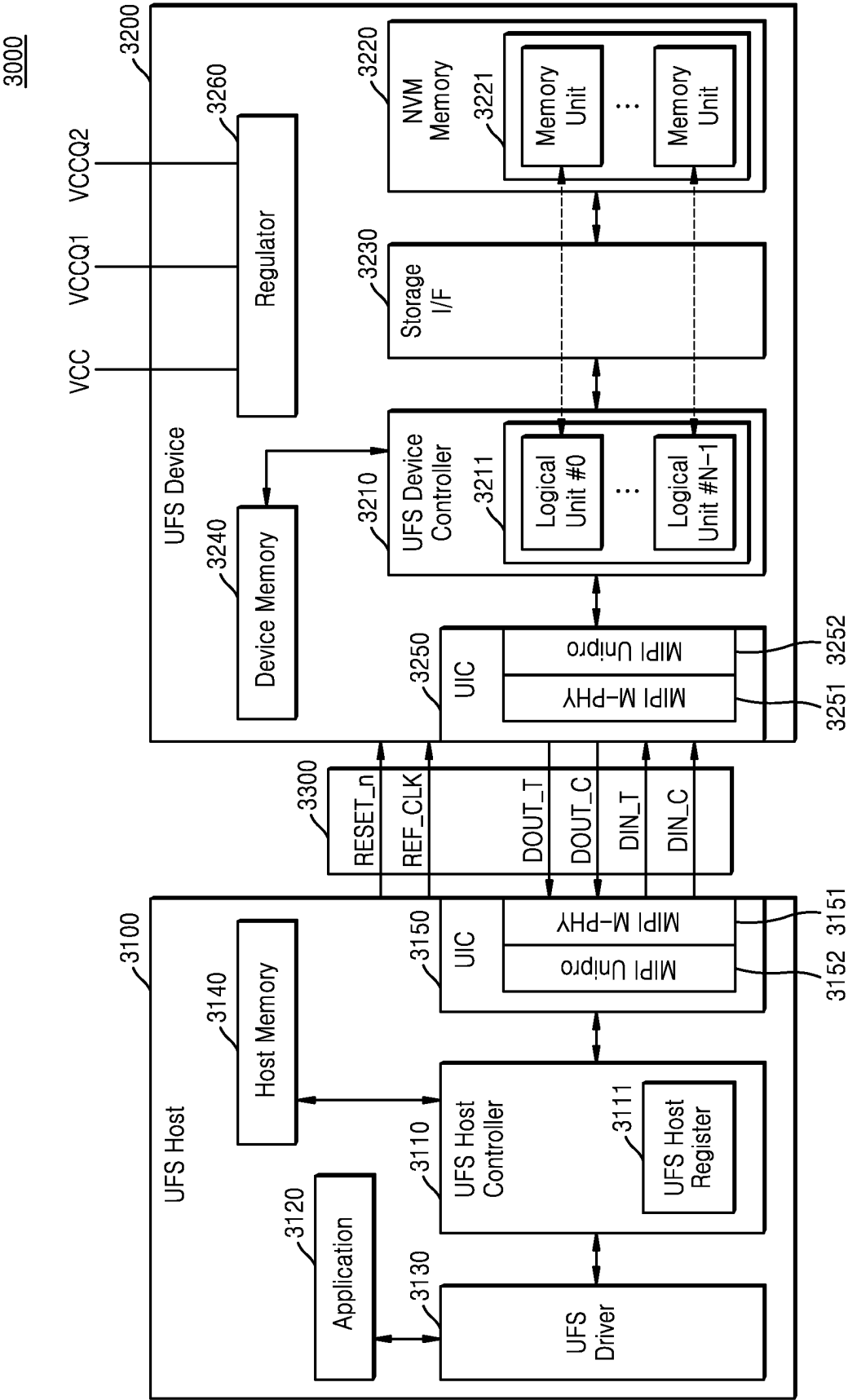




FIG. 16

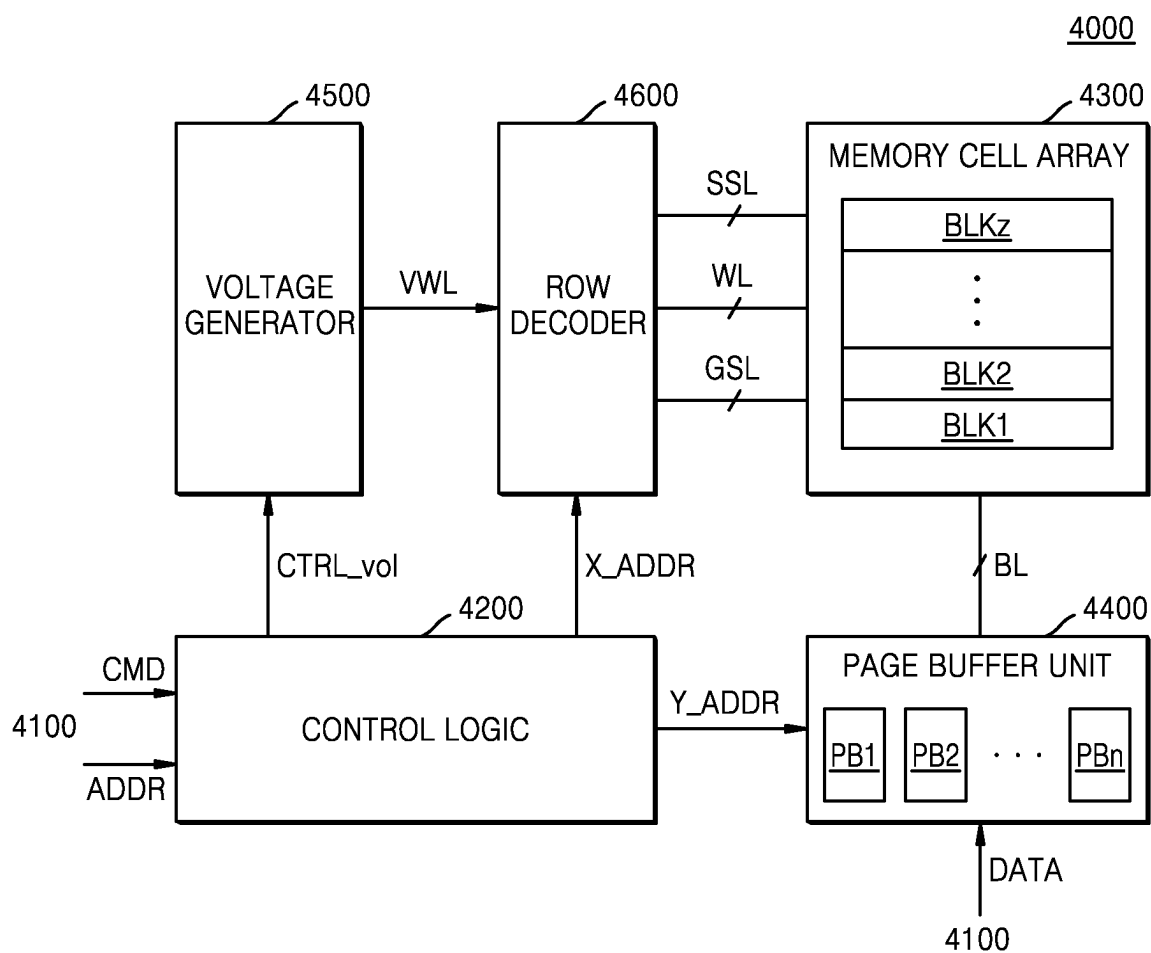
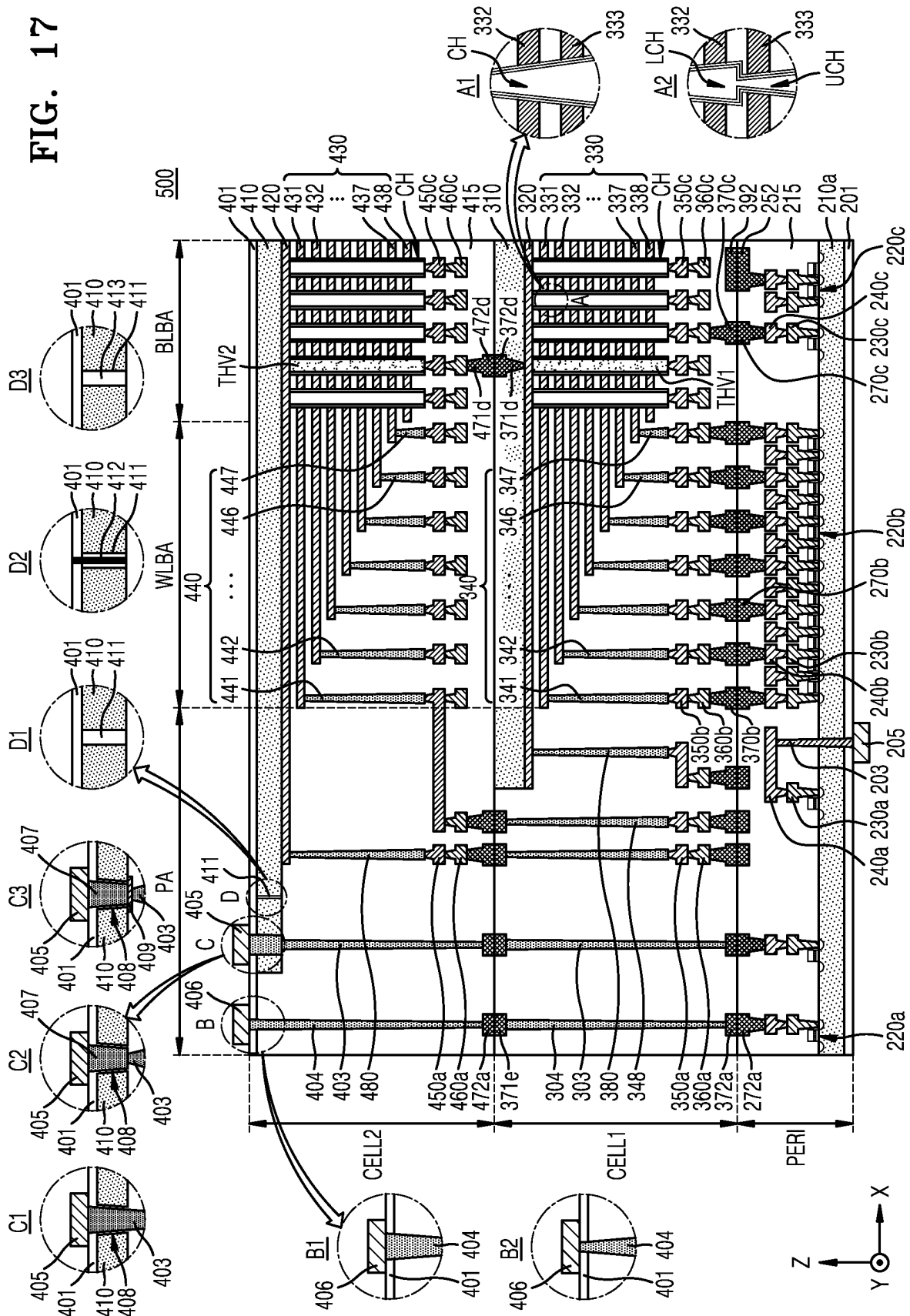


FIG. 17



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# STORAGE DEVICE WITH VARIABLE CELL AREA SIZE AND OPERATING METHOD THEREOF

## CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2023-0017573, filed on Feb. 9, 2023, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

## FIELD

The inventive concept relates to a storage device, and more particularly, to a storage device configured for calculating the number of usable zones based on the size of a variable per-level cell area, and an operating method thereof.

## DISCUSSION

Non-volatile memory may retain stored data even when power is cut off. Storage device formats include flash-based non-volatile memories such as embedded multi-media card (eMMC), universal flash storage (UFS), solid-state drive (SSD), and memory card. Storage devices based on such formats may be usefully employed to store or move data.

A data processing system including a storage device may be referred to as a storage system, and the storage system may include a host and a storage device. The host and the storage device may be connected to each other through various interface standards.

## SUMMARY

An embodiment of the inventive concept provides a storage device that calculates the number of usable zones based on the cell area size of a single-level cell (SLC) area, and allows the host to vary the cell area size of the SLC area, thereby optimizing performance in multiple application operations.

According to an embodiment of the inventive concept, there is provided a storage device including a memory controller and a non-volatile memory, wherein the memory controller is configured to: calculate a number of zones, which are spaces allocated to logical addresses, based on a cell area size of first storage areas of the non-volatile memory, and provide the number of zones to a host; generate multiple zones of a number within the calculated number of zones, and to map a logical address of each of the multiple zones to a physical address of at least one of the first storage areas; and in response to a first write request from the host, write data included in the first write request into first storage areas corresponding to a zone indicated by the first write request.

According to an embodiment of the inventive concept, there is provided an operating method of a storage device including a memory controller and a non-volatile memory including calculating the number of zones, which are spaces allocated to logical addresses, based on cell area sizes of first storage areas of a non-volatile memory, providing a number of zones to a host, generating multiple zones of a number within the calculated number of zones, mapping a logical address of each zone to a physical address of at least one of the first storage areas, and in response to a first write request

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from the host, writing data included in the first write request into first storage areas corresponding to a zone indicated by the first write request.

According to an embodiment of the inventive concept, there is provided a storage system which includes a host and a storage device, wherein the storage device is configured to calculate a number of zones, which are spaces allocated to logical addresses, based on a cell area size of first storage areas of the storage device, and to provide the number of zones to a host; generate multiple zones of a number within the calculated number of zones, and to map a logical address of each zone to a physical address of at least one of the first storage areas; and in response to a first write request from the host, write data included in the first write request into first storage areas corresponding to a zone indicated by the first write request.

## BRIEF DESCRIPTION OF THE DRAWINGS

An embodiment of the inventive concept will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

FIG. 1 is a block diagram illustrating a storage system according to an embodiment;

FIG. 2 is a block diagram illustrating a host according to an embodiment;

FIG. 3 is a block diagram illustrating a storage device according to an embodiment;

FIG. 4 is a conceptual diagram for explaining a non-volatile memory according to an embodiment;

FIG. 5 is a block diagram illustrating the non-volatile memory of FIG. 1;

FIG. 6 is a block diagram illustrating the storage device of FIG. 1;

FIG. 7 is a layout diagram for explaining super block management of a storage device according to an embodiment;

FIG. 8 is a flowchart diagram illustrating an operating method of a storage device according to an embodiment;

FIG. 9 is a block diagram illustrating a storage device according to an embodiment;

FIG. 10 is a data structure diagram for explaining the data structure of a general UPIU according to an embodiment;

FIG. 11 is a data structure diagram illustrating a data structure of a query request UPIU according to an embodiment;

FIG. 12 is a data structure diagram illustrating a data structure of a write command UPIU according to an embodiment;

FIG. 13 is a block diagram illustrating a computing system according to an embodiment;

FIG. 14 is a block diagram illustrating a computing system according to an embodiment;

FIG. 15 is a block diagram for explaining a universal flash storage (UFS) system according to an embodiment;

FIG. 16 is a block diagram illustrating an example of the non-volatile memory of FIG. 1; and

FIG. 17 is a cross-sectional view diagram of a non-volatile memory having a bonding vertical NAND (B-VNAND) structure according to an embodiment.

## DETAILED DESCRIPTION

By way of example, non-limiting embodiments of the present disclosure are described in detail with reference to the accompanying drawings.

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FIG. 1 illustrates a storage system according to an embodiment.

Referring to FIG. 1, a storage system 1 may include a host 10 and a storage device 20.

In an embodiment, the storage device 20 may be implemented as an internal memory embedded into an electronic device, and may be, for example, an embedded universal flash storage (UFS) memory device, an embedded multimedia card (eMMC), or a solid-state drive (SSD). In an embodiment, the storage device 20 may be implemented as an external memory detachable from an electronic device, for example, a UFS memory card, compact flash (CF), secure digital (SD), micro secure digital (Micro-SD), mini secure digital (Mini-SD), extreme digital (xD), or memory stick.

The host 10 may provide logical addresses and commands to the storage device 20. During a write operation, the host 10 may request the storage device 20 to program data to be written into the storage region of the non-volatile memory 220 corresponding to the logical address. During a read operation, the host 10 may request the storage device 20 to sense data to be read from the storage region of the non-volatile memory 220 corresponding to the logical address.

The storage device 20 may include a memory controller 210 connected to a non-volatile memory 220. The memory controller 210 may perform overall control of the storage device 20. Data read from the non-volatile memory 220 may be provided to the host 10, and data provided from the host 10 may be written into the non-volatile memory 220.

The memory controller 210 may control the non-volatile memory 220 to read data stored in the non-volatile memory 220 or to write data to the non-volatile memory 220 in response to a write and/or read request from the host 10.

In detail, the memory controller 210 may control write, read, and erase operations of the non-volatile memory 220 by providing addresses, commands, and control signals to the non-volatile memory 220. In addition, data to be written and data that has been read may be transmitted and received between the memory controller 210 and the non-volatile memory 220.

FIG. 2 illustrates a host 10 according to an embodiment.

Referring to FIG. 2, the host 10 may include a host driver 130, a host memory 100, and a host controller interface 140. The host 10 may be assumed and described as a UFS host conforming to the UFS standard, but embodiments are not limited thereto.

In an embodiment, the host driver 130 may convert I/O requests generated by applications into UFS commands defined by the UFS standard, and transmit the UFS commands to the host controller interface 140. One I/O request may be translated into multiple UFS commands. The I/O requests may also be referred to as task requests. The UFS command may be based on a concept including UFS protocol information units (UPIU) conforming to the UFS standard. The UFS commands may be or include commands defined by a small computer system interface (SCSI) standard, but may also or alternatively be or include commands specific to the UFS standard.

The host controller interface 140 may transmit each UFS command converted by the UFS driver 130 to the storage device 20. In FIG. 2, although the host memory 100 is shown as a separate component from the host controller interface 140; in an embodiment, the host memory 100 may be included in the host controller interface 140. The host controller interface 140 may copy data from a normal region of the host memory 100 to a cache region of the host memory 100 by controlling the host memory 100. The host

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controller interface 140 may transmit a logical address, such as a logical block address (LBA), to the storage device 20.

FIG. 3 illustrates a storage device 20 according to an embodiment. FIG. 3 may be described with reference to FIG. 1.

Referring to FIG. 3, the storage device 20 may include a memory controller 210, a device memory 230 and a non-volatile memory 220. Descriptions of the memory controller 210 and the non-volatile memory 220 have already been provided with reference to FIG. 1, and substantially duplicate description may be omitted.

The device memory 230 may temporarily store data to be written into the non-volatile memory 220 or data that has been read from the non-volatile memory 220. The device memory 230 may include static random-access memory (SRAM) or dynamic random-access memory (DRAM).

The memory controller 210 may calculate the number of zones, as may be described in greater detail below with reference to FIG. 8, based on the cell area size of a single-level cell (SLC) area, without limitation thereto.

According to an embodiment, multi-application performance of the storage device 20 may be optimized as the maximum number of open zones increases.

In detail, although up to two open zones could be provided to the host 10 by calculating the number of open zones based on the buffer of the memory controller 210, but when the maximum number of open zones is calculated based on the size of the SLC region, multiple open zones may be provided to the host 10. Thus, the number of operations that may be simultaneously performed by the storage device 20 may be optimized.

In addition, the memory controller 210 may vary the size of each corresponding SLC area for each data unit to be written depending on the locality and type of data to be written and based on the size of the SLC area included in the request from the host 10, as may be described in greater detail below with reference to FIG. 8.

According to an embodiment, the performance of an SLC write operation for hot data may be optimized by adjusting the size of the SLC area depending on the type and temperature of data. In detail, by making the SLC area corresponding to hot data or warm data greater than the SLC area corresponding to cold data, the performance of the SLC write operation for hot data or warm data may be optimized.

According to an embodiment, the host may change the size of the SLC area, thereby optimizing performance in the multiple application operations.

FIG. 4 illustrates an example for explaining a non-volatile memory 220 according to an embodiment. FIG. 4 may be described with reference to FIG. 3.

Referring to FIG. 4, the non-volatile memory 220 may include a memory cell array 221.

For example, the multiple memory cells included in the memory cell array 221 may be non-volatile memory cells that maintain stored data even when power is cut off. In detail, when the memory cell is a non-volatile memory cell, the non-volatile memory 220 may be an electrically erasable programmable read-only memory (EEPROM), a flash memory, a phase change random-access memory (PRAM), resistance random-access memory (RRAM), a nano floating gate memory (NFGM), a polymer random-access memory (PoRAM), a magnetic random-access memory (MRAM), or a ferroelectric random-access memory (FRAM). Hereinafter, an embodiment of the inventive concept are described taking a case in which multiple memory cells are NAND flash memory cells as an example, but the inventive concept is not limited thereto.

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The memory cell array **221** may include multiple memory blocks, and each memory block may have a planar structure or a three-dimensional structure. The memory cell array **221** may include at least one of a single-level cell (SLC) block including SLCs, a multi-level cell (MLC) block including MLCs, a double-level cell (DLC) block including DLCs, a triple-level cell (TLC) block including TLCs, and a quad-level cell (QLC) block including QLCs, without limitation thereto. For example, some memory blocks among the multiple memory blocks may be the SLC blocks, and other memory blocks may be the DLC blocks, the TLC blocks, and/or the QLC blocks, respectively. In an embodiment, referring to FIG. 4, the memory cell array **221** may include an SLC area **223** that is a set of single-level cell blocks and a TLC area **225** that is a set of triple-level cell blocks.

Each memory cell included in the memory cell array **221** may store data of 2 bits or more. For example, the memory cell may be an MLC that stores 2-bit data. As another example, the memory cell may be a TLC that stores 3-bit data or a QLC that stores 4-bit data. Hereinafter, a memory cell storing data of 2 or more bits is referred to as an MLC. For example, the MLC may include one or more of the DLC, the TLC, and the QLC. In an embodiment, some memory cells included in the memory cell array **221** may be SLCs that store 1-bit data, and other memory cells may be MLCs that store 2-bit data, 3-bit data, 4-bit data, or more bits of data per memory cell.

In an embodiment, the memory cell array **221** may include a three-dimensional (3D) memory cell array, the 3D memory cell array may include multiple NAND strings, and each NAND string may include memory cells respectively connected to word-lines vertically stacked on a substrate. However, the inventive concept is not limited thereto. For example, in an embodiment, the memory cell array **221** may include a two-dimensional (2D) memory cell array.

In addition, the memory cell array **221** may include a first number of memory cell planes including a second number of memory blocks, without limitation thereto. Here, the memory cell array may include a first times second number of memory blocks.

The non-volatile memory **220** may be configured for each memory cell plane to perform a different operation on a memory cell plane basis. That is, the memory cell plane may be a set of memory blocks in which one operation of the non-volatile memory **220** is independently performed. For example, the non-volatile memory **220** may simultaneously perform a read operation from a first memory block of a first memory cell plane and a read operation from a second memory block on a second memory cell plane. This may be referred to as a plane independent read (PIR) or plane independent core (PIC).

In an embodiment, the storage device **20** may perform an SLC buffering operation. The SLC buffering operation represents an operation of preferentially writing data included in a write request into an SLC area among a series of operations of first writing the data included in the write request to the SLC area and later migrating the corresponding data to the MLC region.

Referring to FIG. 4, the memory cell array **221** may include an SLC area **223** that is a set of single-level cell blocks and a TLC area **225** that is a set of triple-level cell blocks.

The storage device **20** may divide the data included in the write request in response to the write request of the host **10**, and may preferentially write the corresponding data into the single-level cell block of the SLC area **223**.

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When a predetermined condition is satisfied, the storage device **20** may migrate data previously written into the single-level cell (SLC) region **223** from the SLC region into the triple-level cell region **225**. When a predetermined condition is satisfied, in response to a write request from the host **10**, a case where the storage device **20** no longer has a single-level cell block available for writing may be included.

FIG. 5 illustrates the non-volatile memory **220** of FIG. 1.

Referring to FIG. 5, a storage region of the non-volatile memory **220** may be configured to include blocks **31**, segments **32**, sections **33**, and zones **34**. Here, the storage region may represent a space allocated in correspondence with a logical address.

The storage region of the non-volatile memory **220** may include multiple zones **34**. Each zone **34** may include multiple sections **33**, each section **33** may include multiple segments **32**, and each segment **32** may include multiple blocks **31**.

For example, the block **31** may be a storage region for storing 4 KB data, and the segment **32** may be a storage region for storing 2 MB data by including 512 blocks **31**. The configuration of the non-volatile memory **220** as shown in FIG. 5 may be determined at the time of formatting the non-volatile memory **220**, but is not limited thereto. Moreover, a section may include two or more segments, and a zone may include two or more sections. The non-volatile memory **220** may read and write data in units of 4 KB pages. That is, one page may be stored in block **31**.

FIG. 6 illustrates the storage device **20** of FIG. 1. FIG. 6 may be described with reference to FIG. 3.

Referring to FIG. 6, the storage device **20** may include a memory controller **210** and a non-volatile memory **220**. The storage device **20** may support multiple channels CH1 to CHm, where m is a natural number, and the non-volatile memory **220** and the memory controller **210** may be connected to each other through the multiple channels CH1 to CHm. For example, the storage device **20** may be implemented as a storage device such as a solid-state drive (SSD).

The non-volatile memory **220** may include multiple non-volatile memory devices NVM11 to NVMmn, where n is a natural number. Each of the non-volatile memory devices NVM11 to NVMmn may be connected to one of the multiple channels CH1 to CHm through a corresponding connection way. For example, the non-volatile memory devices NVM11 to NVM1n may be connected to the first channel CH1 through ways W11 to W1n, and non-volatile memory devices NVM21 to NVM2n may be connected to the second channel CH2 through ways W21 to W2n. In an embodiment, each of the non-volatile memory devices NVM11 to NVMmn may be implemented as an arbitrary memory unit capable of operating based on individual commands from the memory controller **210**. For example, each of the non-volatile memory devices NVM11 to NVMmn may be implemented as a chip or die, but the present invention is not limited thereto.

The memory controller **210** may transmit and receive signals to and from the non-volatile memory **220** through multiple channels CH1 to CHm. For example, the memory controller **210** may transmit commands CMD1 to CMDm, addresses ADDR1 to ADDRm, and data DATA1 to DATAm to the non-volatile memory **220**, or receive data DATA1 to DATAm from the non-volatile memory **220**, through channels CH1 to CHm.

The memory controller **210** may select one of the non-volatile memory devices connected to the corresponding channel through each channel and transmit and/or receive signals to and/or from the selected non-volatile memory

device. For example, the memory controller **210** may select the non-volatile memory device **NVM11** from among the non-volatile memory devices **NVM11** to **NVM1n** connected to the first channel **CH1**. The memory controller **210** may transmit the command **CMD1**, the address **ADDR1**, and the data **DATA1** to the selected non-volatile memory device **NVM11** or may receive data from the selected non-volatile memory device **NVM11**, through the first channel **CH1**.

The memory controller **210** may transmit and receive signals to and from the non-volatile memory **220** in parallel through different channels. For example, the memory controller **210** may transmit the command **CMD2** to the non-volatile memory **220** through the second channel **CH2** while transmitting the command **CMD1** to the non-volatile memory **220** through the first channel **CH1**. For example, the memory controller **210** may receive data **DATA2** from the non-volatile memory **220** through the second channel **CH2** while receiving data **DATA1** from the non-volatile memory **220** through the first channel **CH1**.

The memory controller **210** may control overall operations of the non-volatile memory **220**. The memory controller **210** may control each of the non-volatile memory devices **NVM11** to **NVMmn** connected to the channels **CH1** to **CHm** by transmitting a signal to the channels **CH1** to **CHm**. For example, the memory controller **210** may control a selected one of the non-volatile memory devices **NVM11** to **NVM1n** by transmitting the command **CMD1** and the address **ADDR1** through the first channel **CH1**.

Each of the non-volatile memory devices **NVM11** to **NVMmn** may operate under the control of the memory controller **210**. For example, the non-volatile memory device **NVM11** may program data **DATA1** depending on the command **CMD1**, address **ADDR1**, and data **DATA1** provided through the first channel **CH1**. For example, the non-volatile memory device **NVM21** may read data **DATA2** based on the command **CMD2** and address **ADDR2** provided through the second channel **CH2** and transmit the read data **DATA2** to the memory controller **210**.

FIG. 6 shows that the non-volatile memory **220** communicates with the memory controller **210** through **m** channels and the non-volatile memory **220** includes **n** non-volatile memory devices corresponding to each channel, but the number of channels and the number of non-volatile memory devices connected to each channel may be variously changed, without limitation thereto.

FIG. 7 illustrates an example for explaining super block management of a storage device **20** according to an embodiment.

Referring to FIGS. 4 and 7, the memory cell array **221** may include multiple memory dies, for example, four memory dies **DIE #0** to **DIE #3**. In addition, each memory die may include multiple memory cell planes each including multiple memory blocks, and may include, for example, four memory cell planes **PLANE #0** to **PLANE #3**. The memory controller **210** may divide the memory cell array **221** of the non-volatile memory **220** into multiple super blocks to allocate control the memory cell array **221**. The super blocks may each include two or more memory blocks included in the memory cell array **221**. The super blocks may include multiple physical zones. The super block may be formed across multiple memory dies, and as in FIG. 7, the super block may be formed across substantially all memory dies.

In an embodiment, the super block may be a set of memory blocks having the same block number included in each memory cell plane of substantially all memory dies included in the memory cell array **221**. For example, a first super block **222** may be a set of memory blocks having a

block number of 1, each included in each memory cell plane of substantially all memory dies.

FIG. 8 illustrates an operating method of the storage device **20** according to an embodiment. FIG. 9 illustrates a storage device **20** according to an embodiment.

Referring to FIGS. 3, 5 and 8, in operation **S110**, the storage device **20** may calculate the number of zones, which are spaces allocated to logical addresses, based on the size of the first storage area of the non-volatile memory **220** and provide the number of zones to the host. Here, the first storage area may be an SLC area.

In an embodiment, the memory controller **210** may calculate the number of zones, which may be spaces allocated to logical addresses, based on the size of a super block of SLCs of the non-volatile memory **220**.

When the non-volatile memory **220** includes an SLC area and a TLC area, the memory controller **210** may calculate the number of zones by Equation 1, as set forth below.

$$Num_{Zone} = \frac{\text{Super Block Size for SLC}}{\text{Size for HSP}} \quad [\text{Equation 1}]$$

Here,  $Num_{Zone}$  indicates the maximum number of zones in which simultaneous write operations may be performed in the superblock of the SLC area, Size for HSP represents the size of data required for one write operation in a triple-level cell, and because the value corresponding to the Size for HSP may vary in size depending on the type of multi-level cell used by the memory controller **210**, the inventive concept is not limited thereto. Super Block Size for SLC indicates the size of data within a super block in the SLC area. Here, high speed program (HSP) refers to a program operation of programming multiple data bits to multi-level cells in substantially one operation.

For example, when there are two connection ways, two memory cell planes per way, and four channels, the Super Block Size for SLC may be calculated as 16 KB (Size of SLC page)\*3 (TLC weight)\*4 (CH)\*2 (Die)\*2 (Plane)=768 KB, the Size for HSP may be calculated as 4 (CH)\*2 (Die)\*8 MB (Block size)=128 MB, and the  $Num_{Zone}$  may be calculated as about 168. Here, when calculating the Size for HSP, the HSP operation uses the data unit when HSP is performed in the TLC area.

In an embodiment, in response to the request from the host **10**, the memory controller **210** may calculate the maximum number of zones, which are spaces allocated as logical addresses, based on the number of superblocks to be used as the superblocks of the first storage areas among the multiple superblocks of the non-volatile memory **220** included in the request. Here, the request of the host **10** may be a request of the host **10** to calculate the maximum number of zones. The request of the host **10** may be described in greater detail below with reference to FIGS. 10 and 11.

When the non-volatile memory **220** includes a single-level cell (SLC) area and a triple-level cell (TLC) area, the memory controller **210** may calculate the number of zones by Equation 2 below.

$$Num_{Zone} = \frac{(\text{Super Block Size for SLC}) * (\text{SLC Area Length})}{\text{Size for HSP}} \quad [\text{Equation 2}]$$

Here,  $Num_{Zone}$  indicates the maximum number of zones in which a write operation may be performed simultaneously into the super block of the SLC area, Size for HSP represents

the size of data required for one write operation into a triple-level cell, a value corresponding to the Size for HSP may vary in required size depending on the type of multi-level cell used by the memory controller **210**. Therefore, the inventive concept is not limited thereto. Super Block Size for SLC indicates the size of the super block data in the SLC area, and SLC Area Length is an SLC area length that may be varied by the host **10** and indicates the number of superblocks to be used as superblocks of the SLC area among multiple superblocks. For example, when the SLC Area Length increases by 1, one super block may be further configured as an SLC area.

That is, the memory controller **210** may calculate the number of zones based on the size of the data within the superblock of the SLC area and the number of superblocks to be used as the superblocks of the SLC area among the multiple superblocks

Referring to FIG. 9, the non-volatile memory **220** may include a configurable cell **247**. The configurable cell **247** is an area that may be converted into an SLC area based on the length of the SLC area included in the request of the host. Although one configurable cell **247** is shown in FIG. 9, embodiments are not limited thereto. For example, the configurable cell **247** may be one or multiple configurable cells **247**.

In an embodiment, the memory controller **210** may vary the size of the SLC area based on the configurable cell **247** in response to a request of the host. For example, the memory controller may receive a second write request from the host, and vary the cell area size of the first storage areas for second data from the second write request based on the locality of the second data and/or type of the second data

In addition, the memory controller **210** may provide the number of zones to the host. For example, the memory controller may vary the cell area size of the first storage areas for second data from the second write request based on the size of an area included in the second write request from the host.

In an embodiment, the memory controller **210** may provide the maximum number of open zones to the host **10** based on the zoned block commands (ZBC) or like standard. In this case, the open zone indicates a zone in which a write operation and a read operation may be performed.

In operation **S120**, the storage device **20** may generate multiple zones of a number within the calculated number of zones and may map a logical address of each zone to a physical address of at least one of the first storage areas. In this case, multiple zones may be generated in response to a request of the host **10**.

In an embodiment, the memory controller **210** may generate multiple zones based on the type and locality of data to be written.

Here, the type of data to be written may include data to be written into a data block and a node to be written into a node block corresponding to the data block. The node may include information about a block of data. For example, the node may include a file name, a node identifier, a file size, and/or a logical address of the data block. The name of the file may mean the name of a file including data blocks, the node identifier may be a unique identifier for the node block, and the file size may mean the size of a file including data blocks. Moreover, each node block may refer to multiple data blocks.

In addition, the locality of data to be written may include temporal locality and spatial locality, where temporal locality indicates a property in which an accessed memory space is highly likely to be accessed again in the near future, and

spatial locality indicates a property in which a memory space adjacent to the accessed memory space is highly likely to be accessed. In addition, a locality of data to be written may be referred to as a data temperature.

For example, data to be written based on data temperature may be classified into hot data, warm data, and cold data. Here, data temperatures may be considered in the order of hot data, warm data, and cold data. As used herein, hot data may refer to data that is accessed immediately upon being created, or data that is accessed very frequently. Warm data may refer to data that is accessed within a relatively short period of time after being created, or data that is accessed on a regular basis. Cold data may refer to data that has been created but that is accessed rarely or infrequently.

In addition, data to be written may be classified into hot data, hot node, warm data, warm node, cold data, and cold node based on data temperature and type of data.

In an embodiment, the memory controller **210** may generate a first zone corresponding to the hot data, a second zone corresponding to the hot nodes, a third zone corresponding to the warm data, a fourth zone corresponding to the warm nodes, a fifth zone corresponding to the cold data, and a sixth zone corresponding to the cold nodes. Here, each zone may be or include multiple zones, and for example, the number of first zones corresponding to the hot data may be greater than the number of fifth zones corresponding to the cold data.

In an embodiment, the memory controller **210** may generate zones where sizes of storage regions allocated to the zones corresponding to the hot data are greater than sizes of storage regions allocated to the zones corresponding to the cold data.

In addition, the storage device **20** may map a logical address of each zone to a physical address of at least one of the first storage areas.

In an embodiment, the memory controller **210** may map a logical address of each zone to a physical address of at least one of the SLC areas.

For example, referring to FIG. 9, the memory controller **210** may map one of the SLC areas of the non-volatile memory **220** to correspond to the first zone. Because the first zone corresponds to hot data, the SLC area may be referred to as a hot data SLC region **241**. Similarly, the memory controller **210** may map each of the second to sixth zones to correspond to one of the other SLC areas of the non-volatile memory **220**. The other SLC areas may be referred to as a hot node SLC region **242**, a warm data SLC region **243**, a warm node SLC region **244**, a cold data SLC region **245**, and a cold node SLC region **246**.

Moreover, the storage device **20** may further include a device memory **230** that temporarily stores data to be written into the non-volatile memory **220**.

In an embodiment, the device memory **230** may include buffer regions respectively corresponding to zones generated based on the type and locality of data to be written.

For example, referring to FIG. 9, the device memory **230** may include a hot data region **231**, which is a buffer region corresponding to the first zone, a hot node region **232**, which is a buffer region corresponding to the second zone, a warm data region **233**, which is a buffer region corresponding to the third zone, a warm node region **234**, which is a buffer region corresponding to the fourth zone, a cold data region **235**, which is a buffer region corresponding to the fifth zone, and a cold node region **236**, which is a buffer region corresponding to the sixth zone, without limitation thereto.

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In addition, each of the buffer regions of the device memory **230** may correspond to some of the first storage areas of the non-volatile memory **220**.

For example, the hot data region **231** of the device memory **230** may correspond to the hot data SLC region **241** of the non-volatile memory **220**. In addition, the hot node region **232** of the device memory **230** may correspond to the hot node SLC region **242** of the non-volatile memory **220**. The warm data region **233** of the device memory **230** may correspond to the warm data SLC region **243** of the non-volatile memory **220**. The warm node region **234** of the device memory **230** may correspond to the warm node SLC region **244** of the non-volatile memory **220**. The cold data region **235** of the device memory **230** may correspond to the cold data SLC region **245** of the non-volatile memory **220**. The cold node region **235** of the device memory **230** may correspond to the cold node SLC region **245** of the non-volatile memory **220**.

In operation **S130**, in response to the write request from the host **10**, the storage device **20** may cause data included in the write request to be written into the first storage areas corresponding to the zone indicated by the write request.

The write request of the host **10** may include data to be written, as well as data temperature and type of the data to be written. The write request of the host **10** may be described in greater detail below with reference to FIGS. **10** and **12**.

In an embodiment, the memory controller **210** may write data included in the first write request into the SLC area corresponding to the zone indicated by the first write request in response to the first write request of the host **10**.

For example, when the data temperature of the data to be written is hot and the type of data is node, the memory controller **210** temporarily may store corresponding data into the hot node region **232** of the device memory and write corresponding data to the hot node SLC region **242** of the non-volatile memory **220** corresponding to the hot node region **232**.

In addition, when the storage device **20** receives a write request for a new zone before the storage space of the specific buffer region of the device memory **230** is full, the storage device **20** may write data to be temporarily stored into a specific buffer region into an SLC area of the non-volatile memory **220** corresponding to the specific buffer region.

In an embodiment, the memory controller **210** may be configured to temporarily store data included in the first write request into a first buffer region of the device memory **230** corresponding to the first zone. Thereafter, when the memory controller **210** receives a second write request from the host, including data to be written into a second zone different from the first zone with remaining storage space in the first buffer region, data included in the first write request may be configured to be written into the SLC area corresponding to the first zone in response to the second write request of the host **10**.

For example, it is assumed that the first buffer region and the second buffer region of the device memory **230** are warm data regions, the first buffer region and the second buffer region correspond to the first zone and the second zone, respectively, and the storage spaces of the first buffer region and the second buffer region are 112 KB. The memory controller **210** may temporarily store data included in the first write request of 56 KB into the first buffer region of the device memory **230** corresponding to the first zone. Thereafter, the memory controller **210** may receive a second write request from the host **10**, including data to be written into a second zone different from the first zone. In this case, the

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memory controller **210** may write 56 KB of data temporarily stored in the first zone into the SLC area corresponding to the first zone in response to the second write request from the host **10**.

In addition, the storage device **20** may be configured to migrate data stored in first storage areas corresponding to each zone to second storage areas. Here, the first storage areas may include single-level cells, and the second storage areas may include multi-level cells.

That is, the storage device **20** may perform an SLC buffering operation. The SLC buffering operation indicates an operation of preferentially writing data included in a write request into an SLC area among a series of operations of first writing the data included in the write request to the SLC area and later migrating the corresponding data to the MLC area.

The storage device **20** may divide data included in the write request in response to the write request of the host **10** to preferentially write the corresponding data into the SLC areas **241** to **246**.

When a predetermined condition is satisfied, the storage device **20** may migrate data written into the SLC regions **241** to **246** to the TLC region **248**. A case in which a predetermined condition is satisfied may include a case in which there is no SLC block available to be written into by the storage device **20** in response to a write request of the host **10**. Moreover, another case in which a predetermined condition is satisfied may be where there is fewer than a threshold number of SLC blocks available for writing.

In an embodiment, the memory controller **210** may be configured to migrate data stored in an SLC area corresponding to a specific zone to an MLC region. Here, the MLC region may include a TLC region or a QLC region.

According to an embodiment, the performance of the SLC write operation for the hot data may be optimized by adjusting the size of the SLC area based on the type and temperature of the data. In detail, by making the SLC area corresponding to hot data or warm data greater than the SLC area corresponding to cold data, performance of an SLC write operation for hot data or warm data may be optimized.

According to an embodiment, by allowing the host to adjust the size of the SLC region, predictable data management of the host is possible.

According to an embodiment, as the maximum number of open zones increases, multi-application performance of the storage device may be optimized.

In detail, up to two open zones could be provided to the host by calculating the number of open zones based on the buffer of the memory controller, but when the maximum number of open zones is calculated based on the size of the SLC area, multiple open zones may be provided to the host. In this way, the number of operations that may be simultaneously performed in the storage device may be optimized.

FIG. **10** illustrates an example for explaining the data structure of a general UPIU according to an embodiment.

UFS protocol information units (UPIU) may represent commands or packets transmitted and/or received between a host and a storage device in a UFS device. The UPIU may be or include a command defined by the SCSI standard, but the UPIU may also be a command specific to the UFS standard.

As described in FIG. **8**, a request from the host **10** for calculating the number of zones by the storage device **20** and a write request from the host may correspond to the UPIU.

In detail, referring to FIG. **10**, the general UPIU may have a data structure including a transaction type field, a flags field, a logical unit number (LUN) field, a task tag field,



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initiator ID field, a command set type field, a query function, a task management function field, a response field, a status field, a total extra header segment (EHS) length field, a device information field, a data segment length field, which may hereafter be referred to as a header, transaction specific fields, an extra header segment field, a header end-to-end cyclic redundancy check (E2ECRC) field, a data segment field, and a data E2ECRC field. The length of the header may be 12 bytes, but is not limited thereto. The length of the UPIU may be a minimum of 32 bytes and a maximum of 65600 bytes, but is not limited thereto.

The transaction type field may indicate the type of request or response included in the data structure. In detail, the transaction type may include a transaction code, and the transaction code may define the content, function, or use of the UPIU. The flag field may have different values depending on the transaction type.

The LUN field may include the logical unit number within the target device for which the request is being served. The target device may be, for example, the storage device **20** of FIG. **1**, and a storage region of the storage device **20** may match at least one logical unit number.

The task tag field may be a value corresponding to a task request, and may be a value that increases whenever a new task request is generated. When multiple UPIUs are generated for one task request, substantially all UPIUs may have the same task tag field value. For example, the task tag field may include 8 bits.

The command set type field may indicate a command set type associated with a command UPIU and a response UPIU. The command UPIU may be a UPIU provided by the host **10** to the storage device **20**, and the response UPIU may be a UPIU provided by the storage device **20** to the host **10**.

The query function and task management function field may be a field used in the query request UPIU and query response UPIU to define query functions, and may be a field used in the task management request UPIU to define task management functions.

When a response is requested from the storage device **20**, the response field may indicate success or failure of the requested function.

The status field may include a SCSI status when the UPIU is a response UPIU.

A total EHS length field may indicate the length of an additional header segment in the UPIU. The length of the additional header segment may be a multiple of 4 bytes.

The device information field may provide information about the storage device **20**, that is, device level information, within the response UPIU.

The data segment length field may indicate the number of valid bytes of the data segment in the UPIU.

A transaction specific field may be an additional field required by a specific transaction code.

The additional header segment field may be present when the total EHS length field has a value other than 0, and may have a length corresponding to a multiple of 4 bytes.

The data segment field may have a length corresponding to a multiple of 4 bytes and may include a data payload.

The header E2ECRC field may include CRC data for correcting errors in the header, and the data E2ECRC field may include CRC data for correcting errors in the data segment.

FIG. **11** illustrates a data structure of a query request UPIU according to an embodiment. FIG. **8** may be described with reference to FIG. **10**.

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As described in FIG. **8**, the request of the host **10** to calculate the number of zones by the storage device **20** may correspond to the query request UPIU.

The query request UPIU may be a packet provided by the host **10** to the storage device **20**. Referring to FIGS. **10** and **11**, the transaction code of the query request UPIU may be xx01 0110b, but is not limited thereto.

The query function field of the query request UPIU may have a value representing a query request UPIU for defining a query function.

The SLC region length field of the query request UPIU may have a value for an SLC region length variable by the host **10**. In detail, the SLC region length indicates the number of superblocks to be used as superblocks of the SLC region among multiple superblocks. For example, when the value of the SLC region length increases by 1, one super block may be further configured as the SLC region.

FIG. **12** illustrates a data structure of a write command UPIU according to an embodiment. FIG. **12** may be described in greater detail with reference to FIGS. **8** and **10**.

The write request of the host **10** described in FIG. **8** may correspond to the write command UPIU.

A write request UPIU WRITE COMMAND may be a packet provided by the host **10** to the storage device **20**.

The data temperature field of the write request UPIU may have a value indicating whether data to be written corresponds to hot data, hot node, warm data, warm node, cold data, or cold node. That is, the data temperature field of the write request UPIU WRITE COMMAND includes information about the data temperature and type of data to be written.

FIG. **13** illustrates a computing system **1000** according to an embodiment.

The computing system **1000** of FIG. **13** may be or include a mobile system such as a mobile phone, a smart phone, a tablet personal computer (PC), a wearable device, a health-care device, or an internet of things (IoT) device. However, the computing system **1000** of FIG. **13** is not necessarily limited to a mobile system, and may be a personal computer, a laptop computer, a server, a media player, or an automotive device such as a navigation device. The computing system **1000** of FIG. **13** may include the storage system **1** of FIG. **1**.

Referring to FIG. **13**, the computing system **1000** may include a main processor **1100**, memories **1200a** and **1200b**, and storage devices **1300a** and **1300b**, and may further include one or more of an image capturing device **1410**, a user input device **1420**, a sensor **1430**, a communication device **1440**, a display **1450**, a speaker **1460**, a power supplying device **1470**, and a connecting interface **1480**.

The main processor **1100** may control the overall operation of the computing system **1000**, and more particularly, the operation of other components included in the computing system **1000**. The main processor **1100** may be implemented using a general-purpose processor, a dedicated processor, or an application processor.

The main processor **1100** may include one or more CPU cores **1110** and may further include a controller **1120** for controlling the memories **1200a** and **1200b** and/or the storage devices **1300a** and **1300b**. Depending on an embodiment, the main processor **1100** may further include an accelerator block **1130** that is a dedicated circuit for high-speed data operations, such as artificial intelligence (AI) data operations. Such an accelerator block **1130** may include a graphics processing unit (GPU), a neural processing unit (NPU), and/or a data processing unit (DPU), and may be

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implemented as a separate chip physically independent of other components of the main processor **1100**, without limitation thereto.

The memories **1200a** and **1200b** may be used as main memory devices of the computing system **1000**, may include volatile memories such as SRAM and/or DRAM, and may also include non-volatile memories, such as flash memory, PRAM, and/or RRAM. The memories **1200a** and **1200b** may also be implemented in the same package as the main processor **1100**. The host **10** of FIG. **1** may be implemented using the main processor **1100** and memories **1200a** and **1200b** of FIG. **13**.

The storage devices **1300a** and **1300b** may serve as one or more non-volatile storage devices for storing data regardless of whether power is supplied or not, and may have a relatively large storage capacity compared to the memories **1200a** and **1200b**. The storage devices **1300a** and **1300b** may include storage controllers **1310a** and **1310b** and non-volatile memory (NVM) memories **1320a** and **1320b** that store data under the control of the storage controllers **1310a** and **1310b**. The non-volatile memories **1320a** and **1320b** may include a 2-dimensional (2-D) structure V-NAND flash memory or a 3-dimensional (3D) structure V-NAND flash memory, but may include other types of non-volatile memory, such as PRAM and/or RRAM.

The storage devices **1300a** and **1300b** may be included in the computing system **1000** while being physically separated from the main processor **1100** or may be implemented in the same package as the main processor **1100**. In addition, the storage devices **1300a** and **1300b** may have a form, such as a solid-state device (SSD) or a memory card, and may be detachably coupled to other components of the computing system **1000** through an interface, such as a connection interface as may be described in greater detail below. The storage devices **1300a** and **1300b** may be devices to which standards are applied, such as universal flash storage (UFS), embedded multi-media card (eMMC), or non-volatile memory express (NVMe), but are not necessarily limited thereto. The storage system **1** of FIG. **1** may be included in at least one of the storage devices **1300a** and **1300b** of FIG. **13**.

The photographing device **1410** may capture a still image or a video, and may be a camera, a camcorder, a borescope, a microscope, and/or a webcam.

The user input device **1420** may receive various types of data input from a user of the computing system **1000**, and may be a touch pad, a keypad, a keyboard, a mouse, and/or a microphone.

The sensor **1430** may detect various types of physical quantities that may be obtained from the outside of the computing system **1000**, and may convert the sensed physical quantities into electrical signals. Such a sensor **1430** may be a temperature sensor, a pressure sensor, an illuminance sensor, a position sensor, an acceleration sensor, a biosensor, and/or a gyroscope.

The communication device **1440** may transmit and receive signals between other devices outside the computing system **1000** based on various communication protocols. Such a communication device **1440** may be implemented by including an antenna, a transceiver, and/or a modulator and demodulator (MODEM).

The display **1450** and the speaker **1460** may serve as output devices that output visual information and auditory information to the user of the computing system **1000**, respectively.

The power supply device **1470** may appropriately convert power supplied from a battery and/or an external power

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source connected or built into the computing system **1000** to supply the power to each component of the computing system **1000**.

The connecting interface **1480** may provide a connection between the computing system **1000** and an external device, which, in turn, may be connected to the computing system **1000** and capable of exchanging data with the computing system **1000**.

The connecting interface **1480** may be implemented in various interface methods such as advanced technology attachment (ATA), serial ATA (SATA), external SATA (eSATA), small computer small interface (SCSI), serial attached SCSI (SAS), peripheral component interconnection (PCI), PCI express (PCIe), NVM express (NVMe), IEEE 1394, universal serial bus (USB), secure digital (SD) card, multi-media card (MMC), embedded multi-media card (eMMC), universal flash storage (UFS), embedded universal flash storage (eUFS), and compact flash (CF) card interface.

FIG. **14** illustrates a computing system **2000** according to an embodiment.

The computing system **2000** may include a host **2100** and a storage system **2200**. In addition, the storage system **2200** may include a storage controller **2210** and a non-volatile memory (NVM) **2220**. In addition, according to an embodiment, the host **2100** may include a host controller **2110** and a host memory **2120**. The host memory **2120** may serve as a buffer memory for temporarily storing data to be transmitted to the storage system **2200** or data transmitted from the storage system **2200**. The host **2100** may be an example of the host **10** of FIG. **1**.

The storage system **2200** may include storage media for storing data in response to a request from the host **2100**. For example, the storage system **2200** may include at least one of a solid-state drive (SSD), embedded memory, and removable external memory. When the storage system **2200** is the SSD, the storage system **2200** may be a device conforming to the non-volatile memory express (NVMe) standard. When the storage system **2200** is an embedded memory or an external memory, the storage system **2200** may be a device conforming to a universal flash storage (UFS) or an embedded multi-media card (eMMC) standard. The host **2100** and the storage system **2200** may generate and transmit packets based on each adopted standard protocol.

When a non-volatile memory **2220** of the storage system **2200** includes a flash memory, the flash memory may include a 2D NAND memory array or a 3D or vertical NAND (VNAND) memory array. As another example, the storage system **2200** may include other various types of non-volatile memories. For example, the storage system **2200** may include magnetic RAM (MRAM), spin-transfer torque MRAM, conductive bridging RAM (CBRAM), ferroelectric RAM (FeRAM), phase RAM (PRAM), resistive RAM, and other various types of memories, without limitation.

According to an embodiment, the host controller **2110** and the host memory **2120** may be implemented as separate semiconductor chips. Alternatively, in some an embodiment, the host controller **2110** and the host memory **2120** may be integrated into the same semiconductor chip. For example, the host controller **2110** may be any one of multiple modules included in an application processor, and the application processor may be implemented as a system on chip (SoC). In addition, the host memory **2120** may be an embedded memory included in the application processor, or may be a non-volatile memory or memory module disposed outside the application processor.

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The host controller **2110** may store data, such as but not limited to write data, from the host memory **2120** into the non-volatile memory **2220**, or may manage an operation of storing data, such as but not limited to read data, from the non-volatile memory **220** into the host memory **2120**.

The storage controller **2210** may include a host interface **2211**, a memory interface **2212**, and a central processing unit (CPU) **2213**. In addition, the storage controller **2210** may further include a flash translation layer (FTL) **2214**, a packet manager **2215**, a buffer memory **2216**, an error correction code (ECC) engine **2217** for correctable and/or uncorrectable data errors, and an advanced encryption standard (AES) engine **2218**. The storage controller **2210** may further include a working memory into which the FTL **2214** is loaded, and when the CPU **2211** executes the FTL **2214**, operations of writing and reading data to and from the non-volatile memory may be controlled.

The host interface **2211** may transmit and receive packets to and from the host **2100**.

A packet transmitted from the host **2100** to the host interface **2211** may include a command or data to be written into the non-volatile memory **2220**, and a packet transmitted from the host interface **2211** to the host **2100** may include a response to a command or data read from the non-volatile memory **2220**. The memory interface **2212** may transmit data to be written into the non-volatile memory **2220** to the non-volatile memory **2220** or receive data read from the non-volatile memory **2220**. This memory interface **2212** may be implemented to comply with standard protocols such as Toggle or ONFI.

The FTL **2214** may perform several functions, such as address mapping, wear-leveling, and garbage collection. The address mapping operation is an operation of changing a logical address received from the host **2100** into a physical address used to actually store data in the non-volatile memory **2220**.

The wear-leveling may be a technology for preventing excessive deterioration of a specific block by uniformly using blocks in the non-volatile memory **2220**, and may be exemplarily implemented through a firmware technology that balances erase counts of physical blocks, without limitation thereto. The garbage collection is a technique for securing usable capacity in the non-volatile memory **2220** by copying valid data of a block to a new block and then erasing the old block.

The packet manager **2215** may generate a packet based on the protocol of the interface negotiated with the host **2100** or parse various information from the packet received from the host **2100**. In addition, the buffer memory **2216** may temporarily store data to be written into the non-volatile memory **2220** or data to be read from the non-volatile memory **2220**. The buffer memory **2216** may be included in the storage controller **2210**, but may be disposed outside the storage controller **2210**.

The ECC engine **2217** may perform error detection and correction functions for read data read from the non-volatile memory **2220**. In greater detail, the ECC engine **2217** may generate parity bits for write data to be written into the non-volatile memory **2220**, and the parity bits generated in this way may be stored in the non-volatile memory **2220** together with write data. When data is read from the non-volatile memory **2220**, the ECC engine **2217** may detect and/or correct an error in the read data by using parity bits read from the non-volatile memory **2220** together with the read data, and may output the read data with the error corrected.

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The AES engine **2218** may perform at least one of an encryption operation and a decryption operation on data input to the storage controller **2210** using a symmetric-key algorithm.

FIG. 15 illustrates an example for explaining a UFS system **3000** according to an embodiment

The UFS system **3000** is a system conforming to the UFS standard announced by the Joint Electronics Device Engineering Council (JEDEC), and may include a UFS host **3100**, a UFS device **3200**, and a UFS interface **3300**. The above description of the storage system **1** of FIG. 1 may also be applied to the UFS system **3000** of FIG. 15 within a configurable range that does not conflict with the following description of FIG. 15. That is, the UFS host **3100** of FIG. 15 may include at least some components of the host **10** of FIG. 1, and the UFS device **3200** of FIG. 15 may include at least some components of the storage device **20** of FIG. 1.

Referring to FIG. 15, the UFS host **3100** and the UFS device **3200** may be interconnected through the UFS interface **3300**. When the main processor of the host of FIG. 1 is an application processor, the UFS host **3100** may be implemented as a portion of the corresponding application processor. The UFS host controller **2110** and the host memory **3140** may respectively correspond to a controller and a memory of a main processor of the host **10** of FIG. 1. The UFS device **3200** may correspond to the storage devices **1300a** and **1300b** of FIG. 13, and the UFS device controller **3210** and the UFS device controller **3210** and the non-volatile memory **3220** may respectively correspond to the memory controller **210** and the non-volatile memory **220** of FIG. 1.

The UFS host **3100** may include a UFS host controller **3110**, an application **3120**, a UFS driver **3130**, a host memory **3140**, and a UFS interconnect (UIC) layer **3150**. The UFS device **3200** may include a UFS device controller **3210**, a non-volatile memory **3220**, a storage interface **3230**, a device memory **3240**, a UIC layer **3250**, and a regulator **3260**. The non-volatile memory **3220** may be composed of multiple memory units **3221**, and such a memory unit **3221** may include a 2D structure or a 3D structure V-NAND flash memory, and but may include other types of non-volatile memory, such as PRAM and/or RRAM. The UFS device controller **3210** and the non-volatile memory **3220** may be connected to each other through the storage interface **3230**. The storage interface **3230** may be implemented to comply with standard protocols such as Toggle or ONFI.

The application **3120** may mean a program that wants to communicate with the UFS device **3200** in order to use the functions of the UFS device **3200**. The application **3120** may transmit an input-output request (IOR) to the UFS driver **3130** for input and output (I/O) to the UFS device **3200**. The I/O request (IOR) may mean a read request, a write request, and/or a discard request of data, but is not necessarily limited thereto.

The UFS driver **3130** may manage the UFS host controller **3110** through a host controller interface (UFS-HCI). The UFS driver **3130** may convert I/O requests generated by the application **3120** into UFS commands defined by the UFS standard, and transfer the converted UFS commands to the UFS host controller **3110**. One I/O request may be translated into multiple UFS commands. The UFS commands may be or include commands defined by the SCSI standard, but may also be commands specific to the UFS standard.

The UFS host controller **3110** may transmit the UFS command converted by the UFS driver **3130** to the UIC layer **3250** of the UFS device **3200** through the UIC layer

**3150** and the UFS interface **3300**. In this process, the UFS host register **3111** of the UFS host controller **3110** may serve as a command queue (CQ).

The UIC layer **3150** of the UFS host **3100** may include MIPI M-PHY **3151** and MIPI Unipro **3152**, and the UIC layer **3250** of the UFS device **3200** may similarly include MIPI M-PHY **3251** and MIPI Unipro **3252**.

The UFS interface **3300** may include a line for transmitting a reference clock REF\_CLK, a line for transmitting a hardware reset signal RESET\_n for the UFS device **3200**, a pair of lines for transmitting a pair of differential input signals DIN\_t and DIN\_c, and a pair of lines for transmitting a pair of differential output signals DOUT\_t and DOUT\_c.

For example, the frequency value of the reference clock provided from the UFS host **3100** to the UFS device **3200** may be one of four values of 19.2 MHz, 26 MHz, 38.4 MHz, and 52 MHz, but is not necessarily limited thereto. The UFS host **3100** may change the frequency value of the reference clock even during operation, that is, while data transmission and/or reception is being performed between the UFS host **3100** and the UFS device **3200**. The UFS device **3200** may generate clocks of various frequencies from a reference clock provided from the UFS host **3100** using a phase-locked loop (PLL) or the like. In addition, the UFS host **3100** may set a data rate value between the UFS host **3100** and the UFS device **3200** through the frequency value of the reference clock. That is, the value of the data rate may be determined depending on the frequency value of the reference clock.

The UFS interface **3300** may support multiple lanes, and each lane may be implemented as a differential pair. For example, the UFS interface **3300** may include one or more receive lanes and one or more transmit lanes. In FIG. 15, a pair of lines transmitting a differential input signal pair DIN\_T and DIN\_C may be included in a reception lane, and a pair of lines transmitting a differential output signal pair DOUT\_T and DOUT\_C may be included in a transmission lane, respectively. Although one transmission lane and one reception lane are shown in FIG. 15, the number of transmission lanes and reception lanes may be changed.

The receive lane and the transmit lane may transmit data in a serial communication method, and full-duplex communication between the UFS host **3100** and the UFS device **3200** is possible due to the structure in which the reception lane and the transmission lane are separated. That is, the UFS device **3200** may transmit data to the UFS host **3100** through a transmit lane while receiving data from the UFS host **3100** through a receive lane. In addition, control data, such as commands from the UFS host **3100** to the UFS device **3200** and user data that the UFS host **3100** intends to store in or read from the non-volatile memory **3220** of the UFS device **3200** may be transmitted through the same lane. Accordingly, between the UFS host **3100** and the UFS device **3200**, a separate lane need not be provided for data transmission, other than a pair of reception lanes and a pair of transmission lanes.

The UFS device controller **3210** of the UFS device **3200** may control overall operations of the UFS device **3200**. The UFS device controller **3210** may manage the non-volatile memory **3220** through a logical unit (LU) **3211** that is a logical data storage unit. The number of LUs **3211** may be 8, but is not limited thereto. The LU **3211** may be referred to as a storage region. The UFS device controller **3210** may include an FTL, and may convert a logical data address transmitted from the UFS host **3100**, for example, a logical block address (LBA), into a physical data address, for example, a physical block address (PBA), by using address

mapping information of the FTL. A logical block for storing user data in the UFS system **3000** may have a size within a predetermined range. For example, the minimum size of a logical block may be set to 4 Kbyte.

When a command from the UFS host **3100** is input to the UFS device **3200** through the UIC layer **3250**, the UFS device controller **3210** may perform an operation based on the input command, and when the operation is completed, a completion response may be transmitted to the UFS host **3100**.

As an example, when the UFS host **3100** intends to store user data in the UFS device **3200**, the UFS host **3100** may transmit a data storage command to the UFS device **3200**. Upon receiving a response ready-to-transfer indicating that the user data is ready to be transferred is received from the UFS device **3200**, the UFS host **3100** may transmit the user data to the UFS device **3200**. The UFS device controller **3210** may temporarily store the transmitted user data in the device memory **3240** and store the user data temporarily stored in the device memory **3240** in a selected location of the non-volatile memory **3220** based on the address mapping information of the FTL.

As another example, when the UFS host **3100** intends to read user data stored in the UFS device **3200**, the UFS host **3100** may transmit a data read command to the UFS device **3200**. Upon receiving the command, the UFS device controller **3210** may read user data from the non-volatile memory **3220** based on the data read command and temporarily store the read user data in the device memory **3240**. During this reading process, the UFS device controller **3210** may detect and correct errors in the read user data using a built-in error correction code (ECC) engine. In greater detail, the ECC engine may generate parity bits for write data to be written into the non-volatile memory **3220**, and the parity bits generated in this way may be stored in the non-volatile memory **3220** together with the write data. When data is read from the non-volatile memory **3220**, the ECC engine may correct an error of the read data using parity bits read from the non-volatile memory **3220** together with the read data and output the read data with the error corrected.

In addition, the UFS device controller **3210** may transmit user data temporarily stored in the device memory **3240** to the UFS host **3100**. In addition, the UFS device controller **3210** may further include an advanced encryption standard (AES) engine. The AES engine may perform at least one of an encryption operation and a decryption operation on data input to the UFS device controller **3210** using a symmetric-key algorithm.

The UFS host **3100** may store commands to be transmitted to the UFS device **3200** in order in the UFS host register **3111**, which may serve as a command queue, and transmit commands to the UFS device **3200** in the above order. In this case, the UFS host **3100** may send the next command queued in the command queue to the UFS device **3200** even while the previously sent command is still being processed by the UFS device **3200**, that is, even before the host **3100** receives notification that a previously sent command has been processed by the UFS device **3200**, and thus the UFS device **3200** may also receive the next command from the UFS host **3200** while processing the previously sent command. The maximum number of commands or queue depth that may be stored in such a command queue may be, for example, 32. In addition, the command queue may be implemented as a circular queue type that indicates the beginning and end of a sequence of commands stored in the queue through a head pointer and a tail pointer, respectively.

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Each of the multiple memory units **3221** may include a memory cell array and a control circuit that controls an operation of the memory cell array. The memory cell array may include a 2D memory cell array or a 3D memory cell array. The memory cell array may include multiple memory cells, and each memory cell may be an SLC that stores 1 bit of information, or may be a cell that stores two or more bits of information, such as MLC, TLC, and QLC. The 3D memory cell array may include vertical NAND strings that are vertically oriented where at least one memory cell is located on top of another memory cell.

VCC, VCCQ, VCCQ2, and the like may be input to the UFS device **3200** as power supply voltages. The VCC is a main power supply voltage for the UFS device **3200** and may have a value of about 2.4 V to about 3.6 V. The VCCQ is a power supply voltage for supplying a low-range voltage, mainly for the UFS device controller **3210**, and may have a value of about 1.14 V to about 1.26 V. The VCCQ2 is a power supply voltage for supplying a voltage lower than VCC but higher than VCCQ, is mainly for I/O interfaces, such as MIPI M-PHY **3251**, and may have a value of about 1.7 V to about 1.95 V. The power supply voltages may be supplied to each component of the UFS device **3200** through the regulator **3260**. The regulator **3260** may be implemented as a set of unit regulators each connected to different voltages among the aforementioned power supply voltages.

FIG. 16 illustrates an example of a non-volatile memory **220** of FIG. 1

The non-volatile memory **220** of FIG. 1 may be embodied in a memory device **4000**. Referring to FIG. 16, the memory device **4000** may include a control logic circuit **4200**, a memory cell array **4300**, a page buffer unit **4400**, a voltage generator **4500**, and a row decoder **4600**. The memory device **4000** may further include column logic, a pre-decoder, a temperature sensor, a command decoder, an address decoder, and the like.

The control logic circuit **4200** may overall control various operations within the memory device **4000**. The control logic circuit **4200** may output various control signals in response to the command CMD and/or the address ADDR from the memory interface circuit **4100**. For example, the control logic circuit **4200** may output a voltage control signal CTRL\_vol, a row address X-ADDR, and a column address Y-ADDR.

The memory cell array **4300** may include multiple memory blocks BLK1 to BLKz, where z is a positive integer, and each of the multiple memory blocks BLK1 to BLKz may include multiple memory cells. The memory cell array **4300** may be connected to the page buffer unit **4400** through bit-lines BL, and may be connected to the row decoder **4600** through word-lines WL, string-selection lines SSL, and ground-selection lines GSL.

In an embodiment, the memory cell array **4300** may include a 3D memory cell array, and the 3D memory cell array may include multiple NAND strings. Each NAND string may include memory cells respectively connected to word-lines vertically stacked on a substrate. In an embodiment, the memory cell array **4300** may include a two-dimensional memory cell array, and the 2D memory cell array may include multiple NAND strings disposed in row and column directions.

The page buffer unit **4400** may include multiple page buffers PB1 to PBn, where n is an integer greater than or equal to 3, and the multiple page buffers PB1 to PBn may be respectively connected to memory cells through multiple bit-lines BL. The page buffer unit **4400** may select at least one bit-line from among the bit-lines BL in response to the

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column address Y-ADDR. The page buffer unit **4400** may operate as a write driver or a sense amplifier based on an operation mode. For example, during a program operation, the page buffer unit **4400** may apply a bit-line voltage corresponding to data to be programmed to the selected bit-line. During a read operation, the page buffer unit **4400** may detect data stored in a memory cell by sensing a current or voltage of a selected bit-line.

The voltage generator **4500** may generate various types of voltages for performing program, read, and erase operations based on the voltage control signal CTRL\_vol. For example, the voltage generator **4500** may generate a program voltage, a read voltage, a program verify voltage, an erase voltage, and the like as the word-line voltage VWL.

The row decoder **4600** may select one of the multiple word-lines WL and select one of the multiple string-selection lines SSL in response to the row address X-ADDR. For example, the row decoder **4600** may apply a program voltage and a program verify voltage to the selected word-line during a program operation, and may apply a read voltage to the selected word-line during a read operation.

FIG. 17 illustrates a non-volatile memory **500** having a bonding vertical NAND (B-VNAND) structure according to an embodiment.

Referring to FIG. 17, the non-volatile memory **500** may have a chip to chip (C2C) structure. Here, the C2C structure may refer to fabricating at least one upper chip including the cell region CELL and a lower chip including the peripheral circuit region PERI, respectively, and then connecting the at least one upper chip and the lower chip to each other by a bonding method. For example, the bonding method may refer to a method of electrically or physically connecting a bonding metal pattern formed on an uppermost metal layer of an upper chip and a bonding metal pattern formed on an uppermost metal layer of a lower chip to each other. For example, when the bonding metal patterns are formed of copper (Cu), the bonding method may be a Cu—Cu bonding method. As another example, the bonding metal patterns may also be formed of aluminum (Al) or tungsten (W).

The non-volatile memory **500** may include at least one upper chip including a cell region. For example, as shown in FIG. 17, the non-volatile memory **500** may be implemented to include two upper chips. However, this is an example, and the number of upper chips is not limited thereto. When the non-volatile memory **500** is implemented to include two upper chips, after the first upper chip including the first cell region CELL1, the second upper chip including the second cell region CELL2, and the lower chip including the peripheral circuit region PERI are manufactured, respectively, the non-volatile memory **500** may be manufactured by connecting the first upper chip, the second upper chip, and the lower chip to each other by a bonding method. The first upper chip may be reversed and connected to the lower chip by bonding, and the second upper chip may also be reversed and connected to the first upper chip by bonding. In the following description, upper and lower portions of the first and second upper chips are defined based on before the first upper chip and the second upper chip are reversed. That is, in FIG. 17, the upper part of the lower chip means the upper part defined based on the +Z-axis direction, and the upper part of each of the first and second upper chips means the upper part defined based on the -Z-axis direction. However, this is an example, and alternatively one of the first upper chip or the second upper chip may be reversed and connected by a bonding method.

Each of the peripheral circuit region PERI and the first and second cell regions CELL1 and CELL2 of the non-

volatile memory **500** may include an external pad bonding region PA, a word-line bonding region WLBA, and a bit-line bonding area BLBA.

The peripheral circuit region PERI may include a first substrate **210a** and multiple circuit elements **220a**, **220b**, and **220c** formed on the first substrate **210a**. An interlayer insulating layer **215** including one or more insulating layers may be provided on the multiple circuit elements **220a**, **220b**, and **220c**, and multiple metal wires connected to the multiple circuit elements **220a**, **220b**, and **220c** may be provided in the interlayer insulating layer **215**. For example, the multiple metal wires may include first metal wires **230a**, **230b**, and **230c** connected to each of the multiple circuit elements **220a**, **220b**, and **220c**, and second metal wires **240a**, **240b**, and **240c** formed on the first metal wires **230a**, **230b**, and **230c**. The multiple metal wires may be formed of at least one of various conductive materials. For example, the first metal wires **230a**, **230b**, and **230c** may be formed of tungsten having a relatively high electrical resistivity, and the second metal wires **240a**, **240b**, and **240c** may be formed of copper having a relatively low electrical resistivity.

Although the first metal wires **230a**, **230b**, and **230c** and the second metal wires **240a**, **240b**, and **240c** are shown and described, an embodiment are not limited thereto. At least one additional metal wire may be further formed on the second metal wires **240a**, **240b**, and **240c**. In this case, the second metal wires **240a**, **240b**, and **240c** may be formed of aluminum. In addition, at least some of the additional metal wires formed on the second metal wires **240a**, **240b**, and **240c** may be formed of copper having a lower electrical resistivity than aluminum of the second metal wires **240a**, **240b**, and **240c**.

The interlayer insulating layer **215** may be disposed on the first substrate **210a** and may include an insulating material, such as silicon oxide or silicon nitride.

Each of the first and second cell regions CELL1 and CELL2 may include at least one memory block. The first cell region CELL1 may include the second substrate **310** and the common source line **320**. Multiple word-lines **331** to **338** may be stacked on the second substrate **310** in a direction, such as a Z-axis direction, perpendicular to the upper surface of the second substrate **310**. String-selection lines and ground-selection lines may be disposed above and below the word-lines **330**, and multiple word-lines **330** may be disposed between the string-selection lines and the ground-select line. Similarly, the second cell region CELL2 may include the third substrate **410** and the common source line **420**, and multiple word-lines **431** to **438**, collectively **430**, may be stacked in a direction perpendicular to the upper surface of the third substrate **410**, such as the Z-axis direction. The second substrate **310** and the third substrate **410** may be made of various materials, and may be, for example, a silicon substrate, a silicon-germanium substrate, a germanium substrate, or a monocrystalline epitaxy grown on a monocrystalline silicon substrate. Multiple channel structures CH may be formed in each of the first and second cell regions CELL1 and CELL2.

In an embodiment, as shown in A1, the channel structure CH is provided in the bit-line bonding area BLBA and extends in a direction perpendicular to the upper surface of the second substrate **310** to pass through the word-lines **330**, the string-selection lines, and the ground-selection line. The channel structure CH may include a data storage layer, a channel layer, and a buried insulating layer. The channel layer may be electrically connected to the first metal wiring **350c** and the second metal wiring **360c** in the bit-line bonding area BLBA. For example, the second metal wiring

**360c** may be a bit-line and may be connected to the channel structure CH through the first metal wiring **350c**. The bit-line **360c** may extend in a first direction, such as a Y-axis direction, parallel to the upper surface of the second substrate **310**.

In an embodiment, as shown in A2, the channel structure CH may include a lower channel LCH and an upper channel UCH connected to each other. For example, the channel structure CH may be formed through a process for the lower channel LCH and a process for the upper channel UCH. The lower channel LCH may extend in a direction perpendicular to the upper surface of the second substrate **310** and pass through the common source line **320** and the lower word-lines **331** and **332**. The lower channel LCH may include a data storage layer, a channel layer, and a buried insulating layer, and may be connected to the upper channel UCH. The upper channel UCH may pass through upper word-lines **333** to **338**. The upper channel UCH may include a data storage layer, a channel layer, a buried insulating layer, and the like, and the channel layer of the upper channel UCH may be electrically connected to the first metal wire **350c** and the second metal wire **360c**. As the length of the channel increases, it may be difficult to form a channel having a constant width due to process reasons. The non-volatile memory **500** according to an embodiment may include a channel having optimized width uniformity through a lower channel LCH and an upper channel UCH formed through a sequential process.

As shown in A2, when the channel structure CH is formed to include a lower channel LCH and an upper channel UCH, a word-line positioned near a boundary between the lower channel LCH and the upper channel UCH may be a dummy word-line. For example, the word-line **332** and word-line **333** forming a boundary between the lower channel LCH and the upper channel UCH may be dummy word-lines. In this case, data may not be stored in memory cells connected to the dummy word-line. Alternatively, the number of pages corresponding to memory cells connected to a dummy word-line may be less than the number of pages corresponding to memory cells connected to a general word-line. The voltage level applied to the dummy word-line may be different from the voltage level applied to the general word-line, and thus, the influence of the non-uniform channel width between the lower channel LCH and the upper channel UCH on the operation of the memory device may be reduced.

Moreover, in A2, it is illustrated that the number of lower word-lines **331** and **332** through which the lower channel LCH passes is less than the number of upper word-lines **333** to **338** through which the upper channel UCH passes. However, this is an example, and the present invention is not limited thereto. As another example, the number of lower word-lines through which the lower channel LCH passes may be equal to or greater than the number of upper word-lines through which the upper channel UCH passes. In addition, the structure and connection relationship of the channel structure CH disposed in the first cell region CELL1 described above may be equally applied to the channel structure CH disposed in the second cell region CELL2.

In the bit-line bonding area BLBA, the first through electrode THV1 may be provided in the first cell region CELL1, and the second through electrode THV2 may be provided in the second cell region CELL2. As shown in FIG. 17, the first through electrode THV1 may pass through the common source line **320** and the multiple word-lines **330**. However, this is an example, and the first through electrode THV1 may further penetrate the second substrate **310**. The

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first through electrode THV1 may include a conductive material. Alternatively, the first through electrode THV1 may include a conductive material surrounded by an insulating material. The second through electrode THV2 may also be provided in the same shape and structure as the first through electrode THV1.

In an embodiment, the first through electrode THV1 and the second through electrode THV2 may be electrically connected to each other through a first through metal pattern 372d and a second through metal pattern 472d. The first through metal pattern 372d may be formed on a lower side of the first upper chip including the first cell region CELL1; and the second through metal pattern 472d may be formed on an upper side of the second upper chip, including the second cell region CELL2. The first through electrode THV1 may be electrically connected to the first metal wire 350c and the second metal wire 360c. A lower via 371d may be formed between the first through electrode THV1 and the first through metal pattern 372d, and an upper via 471d may be formed between the second through electrode THV2 and the second through metal pattern 472d. The first through metal pattern 372d and the second through metal pattern 472d may be connected to each other by a bonding method.

In addition, in the bit-line bonding area BLBA, an upper metal pattern 252 may be formed on the uppermost metal layer of the peripheral circuit region PERI, and an upper metal pattern 392 having the same shape as the upper metal pattern 252 may be formed on the uppermost metal layer of the first cell region CELL1. The upper metal pattern 392 of the first cell region CELL1 and the upper metal pattern 252 of the peripheral circuit region PERI may be electrically connected to each other by the bonding method. In the bit-line bonding area BLBA, the bit-line 360c may be electrically connected to a page buffer included in the peripheral circuit region PERI. For example, some of the circuit elements 220c of the peripheral circuit region PERI may provide a page buffer, and the bit-lines 360c may be electrically connected to the circuit elements 220c providing the page buffer through the upper bonding metal 370c of the first cell region CELL1 and the upper bonding metal 270c of the peripheral circuit region PERI.

Continuing to refer to FIG. 17, in the word-line bonding area WLBA, the word-lines 330 of the first cell region CELL1 may extend in a second direction, such as an X-axis direction, parallel to the upper surface of the second substrate 310 and may be connected to multiple cell contact plugs 341 to 347, collectively 340. A first metal wire 350b and a second metal wire 360b may be sequentially connected to upper portions of the cell contact plugs 340 connected to the word-lines 330. The cell contact plugs 340 may be connected to the peripheral circuit region PERI through the upper bonding metal 370b of the first cell region CELL1 and the upper bonding metal 270b of the peripheral circuit region PERI in the word-line bonding area WLBA.

The cell contact plugs 340 may be electrically connected to a row decoder included in the peripheral circuit region PERI. For example, some of the circuit elements 220b of the peripheral circuit region PERI provide a row decoder, and the cell contact plugs 340 may be electrically connected to the circuit elements 220b providing the row decoder through the upper bonding metal 370b of the first cell region CELL1 and the upper bonding metal 270b of the peripheral circuit region PERI. In an embodiment, operating voltages of the circuit elements 220b providing the row decoder may be different from operating voltages of the circuit elements 220c providing the page buffer. For example, operating voltages of the circuit elements 220c providing the page

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buffer may be greater than operating voltages of the circuit elements 220b providing the row decoder.

Similarly, in the word-line bonding region WLBA, the word-lines 430 of the second cell region CELL2 may extend in a second direction, such as the X-axis direction, parallel to the upper surface of the third substrate 410 and may be connected to the multiple cell contact plugs 441 to 447, collectively 440. The cell contact plugs 440 may be connected to the peripheral circuit region PERI through the upper metal pattern of the second cell region CELL2, the lower and upper metal patterns of the first cell region CELL1, and the cell contact plug 348.

In the word-line bonding region WLBA, an upper bonding metal 370b may be formed in the first cell region CELL1 and an upper bonding metal 270b may be formed in the peripheral circuit region PERI. The upper bonding metal 370b of the first cell region CELL1 and the upper bonding metal 270b of the peripheral circuit region PERI may be electrically connected to each other by the bonding method. The upper bonding metal 370b and the upper bonding metal 270b may be formed of aluminum, copper, or tungsten.

In the external pad bonding region PA, a lower metal pattern 371e may be formed in the first cell region CELL1 below the second cell region CELL2, and an upper metal pattern 472a may be formed in the second cell region CELL2. The lower metal pattern 371e of the first cell region CELL1 and the upper metal pattern 472a of the second cell region CELL2 may be connected to each other by the bonding method in the external pad bonding region PA. Similarly, an upper metal pattern 372a may be formed on the first cell region CELL1, and a lower metal pattern 272a may be formed on the peripheral circuit region PERI. The upper metal pattern 372a of the first cell region CELL1 and the lower metal pattern 272a of the peripheral circuit region PERI may be connected to each other by the bonding method.

Common source line contact plugs 380 and 480 may be disposed in the external pad bonding region PA. The common source line contact plugs 380 and 480 may be formed of a conductive material, such as metal, metal compound, or doped polysilicon. The common source line contact plug 380 of the first cell region CELL1 may be electrically connected to the common source line 320, and the common source line contact plug 480 of the second cell region CELL2 may be electrically connected to the common source line 420. A first metal wire 350a and a second metal wire 360a may be sequentially stacked on the common source line contact plug 380 of the first cell region CELL1, and a first metal wire 450a and a second metal wire 460a may be sequentially stacked on the common source line contact plug 480 of the second cell region CELL2.

I/O pads 205, 405, and 406 may be disposed in the external pad bonding region PA. Referring to FIG. 17, a lower insulating layer 201 may cover the lower surface of the first substrate 210a, and first I/O pads 205 may be formed on the lower insulating layer 201. The first I/O pad 205 may be connected to at least one of the multiple circuit elements 220a disposed in the peripheral circuit region PERI through the first I/O contact plug 203 and may be separated from the first substrate 210a by the lower insulating layer 201. In addition, a side insulating layer may be disposed between the first I/O contact plug 203 and the first substrate 210a to electrically separate the first I/O contact plug 203 from the first substrate 210a.

An upper insulating layer 401 covering the upper surface of the third substrate 410 may be formed on the third substrate 410. A second I/O pad 405 and/or a third I/O pad

**406** may be disposed on the upper insulating layer **401**. The second I/O pad **405** may be connected to at least one of the multiple circuit elements **220a** disposed in the peripheral circuit region PERI through the second I/O contact plugs **403** and **303**, and the third I/O pad **406** may be connected to at least one of the multiple circuit elements **220a** disposed in the peripheral circuit region PERI through the third I/O contact plugs **404** and **304**.

In an embodiment, the third substrate **410** may not be disposed in the region where the I/O contact plugs are disposed. For example, as shown in B, the third I/O contact plug **404** may be separated from the third substrate **410** in a direction parallel to the upper surface of the third substrate **410** and may be connected to the third I/O pad **406** through the interlayer insulating layer **415** of the second cell region CELL2. In this case, the third I/O contact plug **404** may be formed through various processes.

Illustratively, as shown in B1, the third I/O contact plug **404** may extend in a third direction, such as the Z-axis direction, and may be formed where a diameter of the third I/O contact plug **404** increases toward the upper insulating layer **401**. That is, while the diameter of the channel structure CH described in A1 is formed to decrease toward the upper insulating layer **401**, the diameter of the third I/O contact plug **404** may be formed to increase toward the upper insulating layer **401**. For example, the third I/O contact plug **404** may be formed after the second cell region CELL2 and the first cell region CELL1 are coupled to each other by the bonding method.

In addition, as shown in B2, the third I/O contact plug **404** may extend in a third direction, such as the Z-axis direction, and may have a smaller diameter toward the upper insulating layer **401**. That is, similar to the channel structure CH, the diameter of the third I/O contact plug **404** may be formed to decrease toward the upper insulating layer **401**. For example, the third I/O contact plug **404** may be formed together with the cell contact plugs **440** before the second cell region CELL2 and the first cell region CELL1 are bonded together.

In an embodiment, the I/O contact plugs may be disposed to overlap the third substrate **410**. For example, as shown in C, the second I/O contact plug **403** is formed to penetrate the interlayer insulating layer **415** of the second cell region CELL2 in a third direction, such as the Z-axis direction, and may be electrically connected to the second I/O pad **405** through the third substrate **410**. In this case, a connection structure between the second I/O contact plug **403** and the second I/O pad **405** may be implemented in various ways.

As shown in C1, an opening **408** penetrating the third substrate **410** is formed, and the second I/O contact plug **403** may be directly connected to the second I/O pad **405** through the opening **408** formed in the third substrate **410**. In this case, as shown in C1, the diameter of the second I/O contact plug **403** may increase toward the second I/O pad **405**. However, this is an example, and the diameter of the second I/O contact plug **403** may be formed to decrease as the second I/O contact plug **403** goes toward the second I/O pad **405**.

As shown in C2, an opening **408** penetrating the third substrate **410** may be formed, and a contact **407** may be formed in the opening **408**. One end of the contact **407** may be connected to the second I/O pad **405** and the other end may be connected to the second I/O contact plug **403**. Accordingly, the second I/O contact plug **403** may be electrically connected to the second I/O pad **405** through the contact **407** in the opening **408**. In this case, as shown in C2, the diameter of the contact may be formed to increase

toward the second I/O pad, and the diameter of the second I/O contact plug may be formed to decrease toward the second I/O pad. For example, the third I/O contact plug **403** may be formed together with the cell contact plugs **440** before the second cell region CELL2 and the first cell region CELL1 are bonded together, and the contact **407** may be formed after the second cell region CELL2 and the first cell region CELL1 are bonded together.

As shown in C3, compared to C2, a stopper **409** may be further formed on the upper surface of the opening **408** of the third substrate **410**. The stopper **409** may be a metal wire formed on the same layer as the common source line **420**. However, this is an example, and the stopper **409** may be a metal wire formed on the same layer as at least one of the word-lines **430**. The second I/O contact plug **403** may be electrically connected to the second I/O pad **405** through the contact **407** and the stopper **409**.

In addition, similar to the second and third I/O contact plugs **403** and **404** of the second cell region CELL2, each of the second and third I/O contact plugs **303** and **304** of the first cell region CELL1 may be formed to have a less diameter toward the lower metal pattern **371e** or may be formed to have a greater diameter toward the lower metal pattern **371e**.

Moreover, according to an embodiment, a slit **411** may be formed in the third substrate **410**. For example, the slit **411** may be formed at an arbitrary position of the external pad bonding region PA. For example, as shown in D, the slit **411** may be located between the second I/O pad **405** and the cell contact plugs **440** when viewed from a plan view. However, this is an example, and the slit **411** may be formed where the second I/O pad **405** is positioned between the slit **411** and the cell contact plugs **440** when viewed from a plan view.

As shown in D2, the slit **411** may be formed to pass through the third substrate **410**. The slit **411** may be used, for example, to prevent the third substrate **410** from being minutely cracked when forming the opening **408**. However, this is an example, and the slit **411** may be formed to a depth of about 60% to about 70% of the thickness of the third substrate **410**.

In addition, as shown in D2, a conductive material **412** may be formed in the slit **411**. The conductive material **412** may be used, for example, to discharge leakage current generated during driving of circuit elements within the external pad bonding region PA to the outside. In this case, the conductive material **412** may be connected to an external ground line.

As shown in D3, an insulating material **413** may be formed in the slit **411**. An insulating material **413** may be formed, for example, to electrically separate the second I/O pad **405** and the second I/O contact plug **403** disposed in the external pad bonding area PA from the word-line bonding area WLBA. By forming the insulating material **413** in the slit **411**, the voltage provided through the second I/O pad **405** may be blocked from affecting the metal layer disposed on the third substrate **410** in the word-line bonding area WLBA.

Moreover, according to an embodiment, the first to third I/O pads **205**, **405**, and **406** may be selectively formed. For example, the non-volatile memory **500** may be implemented to include a first I/O pad **205** disposed on the first substrate **210a**, a second I/O pad **405** disposed on the third substrate **410**, or a third I/O pad **406** disposed on the upper insulating layer **401**.

Furthermore, depending on device criteria and/or design choice, at least one of the second substrate **310** of the first cell region CELL1 and the third substrate **410** of the second



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cell region CELL2 may be used as a sacrificial substrate, and may be substantially completely or partially removed before or after the bonding process. Additional films may be deposited after substrate removal. For example, the second substrate 310 of the first cell region CELL1 may be removed before or after bonding of the peripheral circuit region PERI and the first cell region CELL1, and an insulating layer covering the upper surface of the common source line 320 or a conductive layer for connection may be formed. Similarly, the third substrate 410 of the second cell region CELL2 may be removed before or after bonding of the first cell region CELL1 and the second cell region CELL2, and an upper insulating layer 401 covering the upper surface of the common source line 420 or a conductive layer for connection may be formed.

While the inventive concept has been particularly shown and described by way of example with reference to embodiments thereof, it shall be understood by those of ordinary skill in the pertinent art that various changes in form and details may be made therein without departing from the spirit and scope of the following claims.

What is claimed is:

1. A storage device comprising:  
a memory controller; and  
a non-volatile memory;  
wherein the memory controller is configured to:  
calculate a number of zones, which are spaces allocated to logical addresses, based on a cell area size of first storage areas of the non-volatile memory, and provide the number of zones to a host;  
generate a plurality of zones of a number within the calculated number of zones, and map a logical address of each zone of the plurality of zones to a physical address of at least one of the first storage areas;  
in response to a first write request from the host, write first data included in the first write request into first storage areas corresponding to a zone indicated by the first write request,  
receive a second write request from the host; and  
vary the cell area size of the first storage areas for second data from the second write request based on at least one of the locality of the second data, the type of the second data, or the size of an area included in the second write request from the host.
2. The storage device of claim 1, wherein the memory controller is further configured to:  
generate the plurality of zones based on a type and locality of the first data.
3. The storage device of claim 2, wherein the memory controller is further configured to generate a first zone corresponding to hot data, a second zone corresponding to hot nodes, a third zone corresponding to warm data, a fourth zone corresponding to warm nodes, a fifth zone corresponding to cold data, and a sixth zone corresponding to cold nodes.
4. The storage device of claim 2,  
wherein the storage device further comprises a device memory for temporarily storing data to be written into the non-volatile memory, and  
wherein the device memory is further configured to include buffer areas respectively corresponding to zones generated based on a type and locality of data to be written.
5. The storage device of claim 4, wherein the memory controller is further configured to:

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temporarily store data included in the first write request in a first buffer area of the device memory corresponding to a first zone; and

when a storage space of the first buffer area remains and a second write request including data to be written into a second zone different from the first zone is received from the host, write data included in the first write request into a first storage area corresponding to the first zone, in response to a second write request of the host.

6. The storage device of claim 1, wherein the memory controller is further configured to, in response to the request of the host, calculate the number of zones, which are spaces allocated to logical addresses, based on the number of super blocks to be used as super blocks of the first storage areas among a plurality of super blocks of the non-volatile memory included in the request.

7. The storage device of claim 1, wherein the memory controller is further configured to:  
migrate data stored in the first storage areas to second storage areas.

8. The storage device of claim 7,  
wherein the first storage areas include single-level cells, and  
wherein the second storage areas include multi-level cells.

9. The storage device of claim 2, wherein the memory controller is further configured to generate the plurality of zones where cell area sizes of storage areas allocated to zones corresponding to hot data are greater than cell area sizes of storage areas allocated to zones corresponding to cold data.

10. An operating method of a storage device including a memory controller and a non-volatile memory, the method comprising:

calculating a number of zones, which are spaces allocated to logical addresses, based on cell area sizes of first storage areas of a non-volatile memory;  
providing the number of zones to a host;  
generating a plurality of zones of a number within the calculated number of zones;  
mapping a logical address of each zone to a physical address of at least one of the first storage areas; and  
in response to a first write request from the host, writing data included in the first write request into first storage areas corresponding to a zone indicated by the first write request,

wherein the generating of the plurality of zones includes generating the plurality of zones where cell area sizes of storage areas allocated to zones corresponding to hot data are greater than cell area sizes of storage areas allocated to zones corresponding to cold data.

11. The method of claim 10, wherein the generating of the plurality of zones includes generating the plurality of zones based on a type and a locality of data to be written.

12. The method of claim 10, wherein the generating of the plurality of zones includes generating a first zone corresponding to hot data, a second zone corresponding to hot nodes, a third zone corresponding to warm data, a fourth zone corresponding to warm nodes, a fifth zone corresponding to cold data, and a sixth zone corresponding to cold nodes.

13. The method of claim 11,  
wherein the writing of the data into first storage areas includes temporarily storing data included in the first write request in a first buffer area of a device memory corresponding to a first zone, and

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wherein when a storage space of the first buffer area remains and a second write request from the host including data to be written into a second zone different from the first zone is received, writing data included in the first write request into a first storage area corresponding to the first zone, in response to a second write request from the host.

14. The method of claim 10, wherein the calculating of the number of zones includes calculating the number of zones, which are spaces allocated to logical addresses, based on the number of super blocks to be used as super blocks of the first storage areas among a plurality of super blocks of the non-volatile memory included in the request, in response to the request of the host.

15. The method of claim 10, further comprising:

in response to a second write request from the host, varying the cell area size of the first storage areas for second data from the second write request based on at least one of the locality of the second data, the type of the second data, or the size of an area included in the second write request from the host; and migrating data stored in the first storage areas to second storage areas.

16. The method of claim 15,

wherein the first storage areas include single-level cells, and

wherein the second storage areas include multi-level cells.

17. A storage system comprising a host and a storage device, wherein the storage device is configured to:

calculate a number of zones, which are spaces allocated to logical addresses, based on a cell area size of first

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storage areas of the storage device, and provide the number of zones to a host;

generate a plurality of zones of a number within the calculated number of zones, and map a logical address of each zone to a physical address of at least one of the first storage areas;

in response to a first write request from the host, write data included in the first write request into first storage areas corresponding to a zone indicated by the first write request; and

generate the plurality of zones where cell area sizes of storage areas allocated to zones corresponding to hot data are greater than cell area sizes of storage areas allocated to zones corresponding to cold data.

18. The storage system of claim 17, wherein the storage device is configured to generate the plurality of zones, based on a type and locality of data to be written.

19. The storage system of claim 18, wherein the storage device is configured to generate a first zone corresponding to hot data, a second zone corresponding to hot nodes, a third zone corresponding to warm data, a fourth zone corresponding to warm nodes, a fifth zone corresponding to cold data, and a sixth zone corresponding to cold nodes.

20. The storage system of claim 17, wherein the storage device is further configured to:

receive a second write request from the host; and

vary the cell area size of the first storage areas for second data from the second write request based on at least one of the locality of the second data, the type of the second data, or the size of an area included in the second write request from the host.

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